

SAQA 48872 – Level 5
131 CREDITS - Systems Development

Learner Workbook

National Certificate: Information Technology

Formative Assessment

NATIONAL CERTIFICATE: INFORMATION TECHNOLOGY: SYSTEMS DEVELOPMENT:

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER INFORMATION

Details	Please Complete this Section
Name & Surname:	Simnikiwe Picane
Organisation:	PAICTA NSF
Unit/Dept:	
Facilitator Name:	Nomanesi Dinini
Date Started:	01 April 2026
Date of Completion:	08 July 2026

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WEEK 1

National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

LEARNER WORKBOOK

SAQA ID 114076

Use computer technology to research a computer topic

Unit Std #	114076
NQF Level	4
Notional hours	30
Credit(s)	3
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1 : Plan the research of a computer topic.
• 1. The plan identifies the topic, objectives, and scope of the research. • 2. The plan identifies the time to be taken for the research, the research methods to be used, and the sources of information to be used. • 3. The plan identifies the target audience, presentation methods, and the computer applications to be used for the analysis of data and the presentation of the results of the research.
Specific Outcome 2 : Conduct research of a computer topic using computer technology
• 1. The research conducted accumulates data according to the research plan. • 2. The research conducted provides data analysis with conclusions. • 3. The description of the analysis methods allows the validity of the analysis to be assessed. • 4. Research progress is indicated at intervals by reports, according to the research plan. • 5. The research conducted uses a computer application to analyse the research data.
Specific Outcome 3 : Present the results of research of a computer topic using computer technology
• 1. The presentation is made using the computer application identified in the research plan. • 2. The presentation communicates summarised research data and conclusions to the target audience.

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (3) x 10 = 30	9HRS	21HRS

1.	**Learning Programme Name:**	REFER TO COVER PAGE									
2.	**SAQA Qualification/ Unit Standard Title:**	REFER TO COVER PAGE									
3.	**Qualification/ Unit Standard**		**4.**	**SAQA ID Number**		**5.**	**N Q F Level**	4	**6.**	**Credits**	**3**
7.	**PURPOSE for offering this programme to your learners:**	REFER TO NEXT PAGE									
8.	**TARGET AUDIENCE for this specific programme:**	REFER TO NEXT PAGE									
9.	**Entry/Admission Requirements:**	REFER TO NEXT PAGE									

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 9 hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =21 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				

Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				
4				
5				

Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

2				
3				
4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: Plan the research of a computer topic.
ASSESEMENT CRITERIA
❖ 1. The plan identifies the topic, objectives, and scope of the research. ❖ 2. The plan identifies the time to be taken for the research, the research methods to be used, and the sources of information to be used. ❖ 3. The plan identifies the target audience, presentation methods, and the computer applications to be used for the analysis of data and the presentation of the results of the research.

Activity	Practical	SO	AC
1	**1.1 In pairs outline how you would plan the research of a computer topic (10)**	1	1-3

1.Agree on the mission together

*Nail the question: Turn your broad topic into one specific research quesion.Ex: Not "AI,but "How do transformer models handle long context in 2026-era LLMs?"

*Define done: Decide deliverable and depth.5-min presentation?2000-word report

What's in scope vs out scope?

*Set deadlines:Pick 3 checkpoints: initial sources,draft synthesis,final polish.

2. Split roles, not topic

*Person A: Breadth+ Foundations

.Find core concepts ,key terms,history, landmark papers

.Build glossarry+ timeline

.Sources : textbooks,survey pappers,Wikipedia refers,official docs

Person B: Depth + Current State

*Find implementation, benchmarks, code examples, debates, news

*Look for Github repos, vendor blogs, arXiv,conference talks, state Overflow trends

*Sources : recent papers,product docs,tech Twitter/Threads, changelogs

3.Buids a shared system day 1

*Doc hub: One Google doc/Notion with sections: question| sources| key findings| conflicts| |Next steps

* Source log: Shared table: source| 1-sentence takeaway|Who found it| Reliability 1-5 | Link

*15-min syncs: Every 2 days.Rule: bring 1 new insight + 1 blocker each sync.If stuck,trade sub-questions.

4 Vet and cross-check

*Credibility ladder: Peer-reviwed>vendor whitepapers>engineer blogs>random forum posts.

*Make them connect: Person A's theory should explain person B's examples.If it doesn't, that gap is your next research target.

Track disagreements: Note where sources conflict.Those contraditions usually become your most interesting insights.

5. Synthesize + review as a pair

*Outline together: Decide story flow before writing.

*Write your own half, then swap and edit each other's sections.You'll catch jargon and gaps

*Dry run Q&A: Each person should explain the other's half at a high level.If you can't,it's not ready.

SPECIFIC OUTCOME 2 :

Conduct research of a computer topic using computer technology.

ASSESEMENT CRITERIA

- 1. The research conducted accumulates data according to the research plan.
- 2. The research conducted provides data analysis with conclusions.
- 3. The description of the analysis methods allows the validity of the analysis to be assessed.
- 4. Research progress is indicated at intervals by reports, according to the research plan.
- 5. The research conducted uses a computer application to analyse the research data.

Activity	Practical	SO	AC
2	**1. Describe Data Collection in computer science (6)**	2	1-5

1.Common Data Sources

*APIs(Application Programming Interfaces): Allow

different software application to communicate and transfer data automatically(e.g., fetching live weather data or social media posts).

Sensors and loT (internet of Things): Hardware devices that collect real-world physical data (e.g., temperature, motion, traffic flow).

*Web Scraping: automated bots or scripts used to extract data directly from websites and web pages.

2. Key Methods of capture

*Automated scripts: Custom scripts (frequently written in python) that execute routine, scheduled extraction of data.

*ETL pipelines: extract, Transform, and load(ETL)pipelines pull raw data from various source,clean it, and load it into centralized storage like a Data Warehouse.

*Event Tracking: Monitoring and capturing specific user events (such as "add to cart" or page views) on web and mobile applications.

3. Data Quality Ethics

*Data Validation: Applying rules and constraints at the point of collection to ensure only accurate, formatted, and complete data is ingested.

*Privacy & compliance: Adhering to regulations like the General Data Protection Regulation (GDPR). requires obtaining user consent,security data, and ensuring transparency.

SPECIFIC OUTCOME 3: Present the results of research of a computer topic using computer technology
ASSESEMENT CRITERIA
❖ 1. The presentation is made using the computer application identified in the research plan. ❖ 2. The presentation communicates summarised research data and conclusions to the target audience.

Activity	Question	SO	AC
3	1. **Define the following terms, Coding, Analytical Memos, Displays, Validity, Contextual and Narrative Analysis (10)** 2. **What do you understand by Quantitative content analysis findings (6)**	3	1-2

coding is telling a computer what to do by wring instructions it can understand.

Analytical memos are short, structured documents used by researchers and analysts to capture, track, refine their thoughts during data analysis.They serve as an intellectual diary where you write down reflections, patterns, and evolving theories about your data, rather than just summarizing the facts.

A display is a visual format that systematically organizes and compresses qualitative data to help researchers see patterns, draw conclusions, and take action.

Validity refers to the accuracy, trustworthiness, and credibility of research findings.It ensures that the study's conclusions truly reflect the data and the phenomenon being investigated, rather than researcher bias.

Contextual Analysis is an analytical method that evaluates data by examining the specific environment, background, and cultural circumstances surrounding it.

Narrative Analysis is a research method used method used to analyze stories that people tell during interviws, focus groups, or in texts, focusing on how these stories are structured to make sense of human experiences.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**

Action/s required: **By when:**

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

LEARNER WORKBOOK

SAQA ID 114050

Explain the principles of business and the role of information technology

Learner's name: Simnikiwe

Learner surname: Picane

Id number: 0106181295085

Town: East london

Unit Std #	114050
NQF Level	5
Notional hours	40
Credit(s)	4
Field	Field 10 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences

Standard	

Course:	

Assessor Details				
Name				
Branch			Registration No:	
Contact Details	email:			
	Phone:		Fax:	

Moderator Details				
Name				
Branch				
Contact Details	email:		Registration No:	
	Phone:		Fax:	

Candidate Details				
Name			ID No	
Job			Company	
Department				
Contact Details	Email:			
	Phone:		Fax:	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		Date	
Assessor's Signature		Date	
Moderator's Signature		Date	

Control sheet

Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				
4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



Specific Outcome 1 : Describe fundamental business concepts.
• 1. The description distinguishes types of business organisations. • 2. The description outlines the common objectives within which businesses operate. • 3. The description outlines the environment within which businesses operate..
Specific Outcome 2 : Describe systems theory with respect to information systems.
• 1. The description defines the concept of a system. • 2. The description identifies theoretical components of a system. • 3. The description distinguishes types of information systems.
Specific Outcome 3 : Explain how IT can be used in business.
• 1. The explanation identifies the purpose of computer applications in business. • 2. The explanation outlines the functions of computer applications in business. • 3. The explanation illustrates the effects of IT on business systems.
Specific Outcome 4 : Explain the relationship between a business and its information needs.
• 1. The explanation distinguishes data and information. • 2. The explanation outlines the role of information in decision making. • 3. The explanation identifies the main threats to data security and integrity. • 4. The explanation identifies the sub-systems that make up a business and the information needs associated with each sub-system.

SPECIFIC OUTCOME 1: Describe fundamental business concepts.
ASSESEMENT CRITERIA
❖ 1. The description distinguishes types of business organisations. ❖ 2. The description outlines the common objectives within which businesses operate. ❖ 3. The description outlines the environment within which businesses operate..

Activity	Question	SO	AC
1	1. **List advantages and disadvantages of each form of business organization** 2. **List Types of Business Environment** 3. **Distinguish between internal and External Environment**	1	1-3

1.sole proprietorship

. Advantages:

*Easy setup: Minimal legal paperwork and very low startup costs.

*Complete control: The owner makes all management decisions independently.

*Tax simplicity: Profits are taxed once as personal income (pass-through taxation).

*Direct retention: The owner keeps 100% of the generated profits.

.Disadvantages:

*Unlimited liability: The owner is personally responsible for all business debts and lawsuits.

Limited capital: Raising money is difficult as you cannot sell business shares.

*Lack of continuity: The business legally terminates if the owner dies or retires

*Heavy workload: The owner must handle all bussiness operations alone.

—

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• 1. The description distinguishes types of business organisations.			
• 2. The description outlines the common objectives within which businesses operate.			

SPECIFIC OUTCOME 2 : Describe systems theory with respect to information systems.
ASSESEMENT CRITERIA
❖ 1. The description defines the concept of a system. ❖ 2. The description identifies theoretical components of a system. ❖ 3. The description distinguishes types of information systems.

Activity	Question	SO	AC
2	**1. What is the importance of Defining information systems logically** **2. List the hierarchical structure of an information system**	2	1-3

Focuses on True Business Needs

*Prevents technical distraction: It keeps the focus on user requirements and business

processes rather than getting bogged down in hardware specs or coding languages.

*Ensures the right problem is solved:

Developers design what the business actually needs, rather than what

a specific software tool limits them to doing.

2.Provides Technology Independence

*Future-proofs the design: Because a logical design does not mention specific brands or vendors, the underlying technology can change without changing the core business logic.

*Easier migrations: If a company decides to switch from an on-premise server to the cloud, the logical blueprint remains exactly the same.

3.Enhances Communication with Non-Technical Stakeholders

*Universal language:Business managers, clients, and end-users can easily understand logical models (like Data flow Diagrams or use cases) because they lack confusing tecnical jargon.

*Easily error detection : stakeholders can spot missing requirements or incorrect business logic before expensive coding or construction begins.

—

4.Maximizes Design Flexibility

*Allows for multiple physical options: A single logical design can be implemented in many different ways (e.g., built from scratch , bought as off-the-shelf software, or outsourced).

*Reduces redesign costs: Making a change on a logical diagram costs virtually nothing compared to rewriting databases or altering physical network layouts later in the project.

5.Improves System Maintenance and Documentation

*clearer troubleshooting: when a system breaks or needs an upgrade, technical terms

*Easier developer onboarding: New engineers can quickly grasp how data moves through business.

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• 1. The description defines the concept of a system.			
• 2. The description identifies theoretical components of a system.			
3. The description distinguishes types of information systems.			

A system is a collection of interrelated and interdependent components that work together within a defined boundary to achieve a common purpose or goal.

SPECIFIC OUTCOME 3: Explain how IT can be used in business.
ASSESEMENT CRITERIA
❖ 1. The explanation identifies the purpose of computer applications in business. ❖ 2. The explanation outlines the functions of computer applications in business. ❖ 3. The explanation illustrates the effects of IT on business systems.

Activity	Question	SO	AC
3	**1.1 Explain how IT can be used in business.**	3	1-3

Communication and Collaboration

Business use technology to facilitate seamless communication among employees, clients, and patners.

Tools such as email, instant messaging, video conferencing, and cloud-based collaboration

platforms allow teams to work together efficiently, regardless of location. This reduces delays,

lowers costs, and improves productivity by enabling real-time data sharing and project management tracking.

Data Management and Business Intelligence

IT systems allow businesses to collect, store, and analyze large volumes of data, including customer

information, sales trends, and market insights. This data can inform strategic decisions, optimize

operations, and identity new opportunities. Business intelligence tools help organizations visualize

data, track performance metrics, and make evidence-based decisions.

Productivity and Workflow Optimization

Software applications for word processing, spreadsheets, project management, and task allocation streamline daily operations. Tools like Microsoft Office, Asana, and Trello help employees organize work, monitor progress, and collaborate effectively. Automation through IT reduces repetitive tasks, saving time and minimizing errors.

Customer Relationship Management (CRM)

CRM system such as salesforce or HubSpot enable businesses to manage customer interactions, track sales processes, and provide personalized support. By analyzing customer behavior and preferences, companies can improve service quality, enhance, customer satisfaction, and build long-term loyalty.

Conclusion

By integrating technology into communication, data management, productivity, customer relations, e-commerce,supply chain, and security, businesses can operate more efficiently,make informed decisions, and remain competitive in a evolving market.

Self-check

OUTCOME	**Yes**	**No**	**I Need help**
• 1. The explanation identifies the purpose of computer applications in business.			
• 2. The explanation outlines the functions of computer applications in business.			
• 3. The explanation illustrates the effects of IT on business systems.			

SPECIFIC OUTCOME 4: Explain the relationship between a business and its information needs.
ASSESEMENT CRITERIA
❖ 1. The explanation distinguishes data and information. ❖ 2. The explanation outlines the role of information in decision making. ❖ 3. The explanation identifies the main threats to data security and integrity. ❖ 4. The explanation identifies the sub-systems that make up a business and the information needs associated with each sub-system.

Activity	Question	SO	AC
4	**1. Describe the Role of information in decision making** **2. Define a system**	4	3

Clarity and Informed Choices

Information provides clarity and understanding about the situation at hand,allowing decision-makers to comprehend the context, implications, and potential outcomes of different options. Access to accurate and relevant data ensures that decisions are fact-based rather than assumption-driven, reducing errors and improving the likelihood of successful outcomes in both personal and professional settings. For example, individuals may rely on health data or financial reports to make critical life choices, while businesses use market research and consumer feedback to guide strategy decisions.

Risk Assessment

Is to identifying and evaluating risks by analysing data, decision-makers can anticipate potential threats.This ensbles strategic risk management,helping organise and minimize potential harm while maximizing benefits.

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• 1. The explanation distinguishes data and information.			
• 2. The explanation outlines the role of information in decision making.			
• 3. The explanation identifies the main threats to data security and integrity.			

Conclusion

In summary, information is central to decisions-making, influencing clarity, risk assessment, strategic planning, problem-solving, and innovation. The effectiveness of decisions depends not only on the availability of information but also on the ability to interpret and apply it effectively. Proper management of information ensures that decisions are informed, strategic, and capable of achieving optimal outcomes across all areas of life.

2.Information is a fundamental element in decision-making, serving as the foundation for informed choices. It plays a crucial role in various aspects of decision -making,including problem-

solving, evidence-based decision-making, risk assessment, forecasting, and improving accuracy.

Information helps identify problems, evaluate alternatives, and make informed decisions that minimize potential harm and maximize benefits. It is also a catalyst for innovation, driving new ideas, products, and services through the accumulation and analysis of existing knowledge.

The quality, quantity, and accessibility of information significantly influence outcomes, whether in personal decision-making or within professional contexts.

3.The main threats to data security and integrity include malware, phishing attacks, vulnerabilities, backdoor attacks, formjacking, cryptojacking, DDoS attacks, and DNS poisoning attacks.Malware refers to software designed to disrupt, damage, or gain unauthorized access to computer systems or sensitive information.

Look back at what you have learnt in this unit standard. Make any suggestions or comments that need to be considered.

Reflection No:	**1**		
Date started:		**Date completed:**	

1. Did this learning guide help you to develop or revive new skills?	Yes	No
2. Are the materials and training you received clear and easy to understand?	Yes	No
3. Did the mentor explain to you the purpose of this training?	Yes	No
4. Do you find your facilitator to be supportive, dedicated and motivated to develop you?	Yes	No
5. Do you enjoy the learning programme?	Yes	No
6. Are you still motivated to participate in this learning programme?	Yes	No
7. Is there sufficient time to cover your specific problem areas in detail?	Yes	No
8. Which information/material do you find most helpful/useful?		
9. Which specific skills and knowledge did you gain from this training?		
10. Do you have any suggestions or comments about the course that will help us to improve future training?		

I have improved

Very much	More than I expected	As I expected	Not as much as I expected	Not at all

I rate my overall experience of the learning guide as

Excellent	Good	Average	Below average	Poor

Thank you for your comments and valuable participation

Application for Assessment

First Attempt □ Second Attempt □ Third Attempt□ RPL □

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time		
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	Theory	Practical
Date of assessment:		
When would be the best time for the Assessment?		

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments 4. Conduct observation in workplace where applicable 5. Complete assessor toolkit

WEEK 2

National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

LEARNER WORKBOOK

SAQA: 116389

WRITE A TECHNICAL REPORT

Learner Information:

Details	Please Complete this Section
Name & Surname:	Simnikiwe Picane
Organisation:	PAICTA NSF
Unit/Dept:	
Facilitator Name:	Nomanesi Dinini
Date Started:	01 April 2026
Date of Completion:	08 July 2026

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OVERVIEW			
About the Learner Workbook	This Learner Exercise Workbook has been designed and developed to evaluate learners' level of understanding of the **WRITE A TECHNICAL REPORT** It forms part of a series of Learner Workbooks that have been developed for **National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS**		
Purpose	The purpose of this Learner Exercise Workbook is to evaluate learners understanding on the specific outcomes and/or assessment criteria of the following SAQA Registered Unit Standards:		
US No	**US Title**	**Level**	**Credits**
116389	**WRITE A TECHNICAL REPORT**	**4**	**4**
Context	This assessment represents the Formative Assessment component of Function in a business environment, for the **National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS** nd should be completed in the classroom/training room.		
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.		
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the workbook to each learner; and • Interview the learner on similar questions, should he/ she not be able to write.		

Instructions to Learners	Learners will be required to: • Complete the workbook as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame of.... hours.
Total Mark	This formative assessment carries a total mark of _ **points**. In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.
Equipment	Learners are required to have the following equipment in order to complete this workbook: • Pen and Pencil; • Ruler; and • Exam Pad – for additional paper.

1. GENERAL INFORMATION

LEARNER DETAILS		
Learner Full Names	Simnikiwe Picane	
Learner ID No.:	0106181295085	
Organisation:	PAICTA NSF	
Unit/Dept:		
Contact Details:	**Telephone /Cell Numbers:**	**Email Address:**
	0710247715	Simnikiwepicane06@gmail.com
WORKSHOP DETAILS		
Workshop Venue:		
Facilitator Name:		
Date Started:		
Date Completed:		

2. ASSESSMENT PREPARATION CHECKLIST

DESCRIPTION	YE	N	COMMENTS/
This assessment is a formative assessment and it is based on the outlined unit standard/s for **WRITE A TECHNICAL REPORT** module.			
Your assessment evidence for **WRITE A TECHNICAL REPORT** module needs to be submitted on..... (day) of..... (month).....(year) at the following address/			
You will be assessed based on the outlined Unit Standards. The assessment activities are linked to specific outcomes/assessment criteria of the outlined Unit Standards.			
To determine your competence level, the following are the methods to be used for this assessment: 1.			
To be declared competent on **WRITE A TECHNICAL REPORT** module (formative assessment), you should have obtained at least 80% of the total mark of this assessment.			
You will be provided with detailed feedback on your performance of this assessment as follows: 1. Written Feedback			
Should you be declared "not yet competent" on this assessment, you will be entitled for re-assessment opportunity/ies.			
You will be required to re-submit evidence (only for areas) you were declared not yet competent. A date for re-submission will be			

You will be entitled to lodge an appeal should you not be satisfied with the assessment decision of your assessment.			
You will be required to provide the assessor feedback on assessment procedure – this is to assist in improving the assessment practices.			
Your results of assessment and portfolio of evidence information will not be provided to any person without your written consent.			

Learner's Declaration			
I Simnikiwe Picane.....herewith declare that I am ready for the assessment, that we have reviewed the assessment preparation and plan, I understand the assessment process and I am happy that the assessment will be conducted in a fair manner.			
Learner Signature:	**Date:**	**Facilitator**	**Date:**
S.Picane	08 July 2026		

Learning Unit 1 Write a technical report

UNIT STANDARD NUMBER	:	116389
LEVEL ON THE NQF	:	4
CREDITS	:	4
FIELD	:	Manufacturing, Engineering and Technology
SUB FIELD	:	Manufacturing and Assembly

PURPOSE:	• Write a clear, complete and correct technical report, which contains all the necessary information and which is easy to understand.
LEARNING ASSUMED TO BE IN PLACE:	
Sufficient knowledge of the subject of the report and know how to interpret	

SESSION 1.

Collect information for writing the report.

Learning Outcomes

- The purpose of the report is determined and clearly defined.
- All facts, which have a bearing on the report are ascertained.
- The correct equipment and/or tools required to ascertain the facts is selected.
- Those facts to be presented in the report are selected and put in the order of presentation.
- The order of presentation is logical.
- The consequences of providing incorrect information are listed and explained.

Activity	Questions Description	Mark
1	**What is the purpose of a report**	4

A report is a formal document that presents information in an organized format for a specific audience and purpose. It is typically a concise piece of wrtting that uses facts and evidence to analyze issues, situations, events, or findings. Reports can summarize facts on a specific topic, issue, or may also provide detailed analyses of ongoing or completed research, initiatives, or projects.

Activity	Questions Description	Mark
2	**Explain the consequences of providing incorrect information in a report**	3

Risks of Inaccurate Financial Reporting

Inaccurate financial reporting can lead to a range of negative consequences, including poor business and investment decisions, regulatory fines, reputational damage, economic loss, penalties, fines, legal action, and even bankruptcy. Companies can mitigate these risks by employing internal controls, financial software, and regular audits to ensure accurate reporting.

Activity	Questions Description	Mark
3	**List ways to Collect information for writing the report.**	5

Effective Data Collection Methods

When it comes to collecting information for writing a report, there are several effective methods to consider. Here are some of the best ways to gather information:

*Surveys and Questionnaires: Design and distribute surveys or questionnaires to gather data from a large of group of people

*Interviews: Conduct one-on-one or group interviews with relevant stakeholders to gather in-depth information.

*Observation: Observe and document the subject of the report to gather first-hand information

*Research: Utilize academic journals, books, articles, and credible online source to gather relevant data and information.

*Data Analysis: Analyze existing data sets, statistics,and reports to exatract relevant information for the report.

*Experiments and Tests: Conduct experiments or tests to gather specific data or evidence for the report.

*Case Studies: Analyze real-life cases or scenarios to gather insights and information relevant to the report.

*Focus Groups: Organize focus group discussions to gather opinions and insights from a diverse group of individuals.

*Document Review: Review relevant documents, reports, and records to gather information for the report.

*Internet Research: Utilize online resources, databases, and credible websites to gather information relevant to the report.

These methods can help you gather a wide range of information, from quantitave data to

SESSION 2.

Plan the writing of the report.

Learning Outcomes

- The order of presentation of the facts is determined.
- Appropriate headings are selected for the report.
- Only those facts, which are necessary for the report to achieve its purpose are selected.
- All facts and figures are put together in a way the reader will be able to understand.

Activity	Questions Description	Mark
4	**Describe the general format of a report**	10

Report Writing Format

*Title Page: Report Title, Author(s), date, and Organisation (if applicable).

Table of Contents: A list of sections and their corresponding page numbers for easy navigation.

*Executive Summary or Abstract: A brief overview of the report's purpose, findings,conclusions, and recommendations.

*Introduction: Background information, context, purpose, objectives, scope, and limitations of the report.

*Methodology: Description of the methods used to gather data or conduct research.

*Findings or Results: Presentation of the data collected or the result of the research.

*Discussion or Analysis: Interpretation of the findings, discussion of implication implications, significance, and any patterns or trends observed.

Conclusion: Summary of the report's findings and recommendations

*Appendices: Additional information, data, or materials that support the report's findings.

This structure ensures clarity, coherence, and impact, making it easier for readers to understand and act on the information presented.

SESSION 3.

Write the report.

Learning Outcomes

- The wording and phraseology is appropriate for the recipient.
- The layout of the report is correct.
- The report contains the correct headings.
- The report is unbiased and factual.
- The report is readable.

Activity	Questions Description	Mark
5	**In groups search for different types of technincal reports and comment on the following :** • **wording and phraseology.** • **layout of the report.** • **headings.** • **unbiased and factual.** • **the report is readable.**	**10**

Technical reports are highly structured, objective documents designed to communicate complex information clearly and efficiently to a specific audience. When analyzing different types of technical documents—such as Scientific Reports, Engineering Reports, and Business/Feasibility Reports—their components can be evaluated against the five requested core criteria:

. Wording and Phraseology Formal Tone: Authors write in an objective, impersonal, and professional style

Active vs. Passive Voice: Engineering and scientific reports historically favored the passive voice ("An

was conducted"), but modern standards increasingly prefer the active voice ("We conducted the experiment") for clarity.

Domain Jargon: Jargon is used precisely when addressing technical peers, but it is minimized or explicitly

defined in the introduction or glossary when writing for mixed audiences.

Conciseness: The phraseology prioritizes economy of language, avoiding fluff, flowery adjectives, of

~~emotional~~ modifiers

Layout of the Report

Front Matter: Includes a Title Page, an Abstract or Executive Summary, and a Table of Contents to navigate

dense content.

The Body: Structured logically using standardized frameworks. For example, scientific reports follow the

IMRaD format (Introduction, Methods, Results, and Discussion).

Back Matter: Contains lists of references, bibliographies, and appendices for detailed raw data, calculations,

or maps.

Visual Integration: Data is strategically integrated using labelled tables, graphs, and schematics rather than

~~relying solely on large~~ blocks

Headings

Signposting: Headings and subheadings act as visual signposts that let readers scan the document and jump directly to relevant data

Hierarchical Numbering: Standard technical layouts utilize a decimal numbering system (e.g., Heading 1.0, ~~Subheading 1.1, Sub-subheading 1.1.1) to clearly~~

Descriptive Titles: Instead of generic headers, effective reports use descriptive phrases (e.g., "4.0 Lithium-Ion Battery Thermal Degradation Results") to prepare the reader for the exact data that follows

. Unbiased and Factual

Evidence-Based: Every claim, assertion, or conclusion is strictly backed by empirical data, mathematical calculations, citations, or physical testing.

Neutral Language: The language avoids personal insights, emotional biases, or unsubstantiated speculation.

Error Acknowledgment: Unbiased technical writing openly notes data limitations, margins of error, and potential anomalies in the testing environment rather than hiding them.

. Readability

Structural Breakout: Density is minimized by leveraging short paragraphs, explicit bullet points, and numbered lists.

Typography: The text uses clean, standard sans-serif fonts (like Arial or Calibri) with consistent line spacing, clear margins, and distinct text sizing for visual comfort.

Audience Adaptation: High readability relies heavily on matching the reader's technical literacy. Complex technical concepts are paired with analogies or diagrams when targeted at executive decision-makers.

SESSION 4.

Revise the report.

Learning Outcomes

- The report is checked to ensure it is clear, concise, complete and correct.
- The report is written or typed.

Activity	Questions Description	Mark
6	**What is the important of revising a report**	4

The revision process sharpens your work across several key areas:

Improved Clarity: Helps you organize ideas logically so readers can easily grasp your findings and conclusions.

Enhanced Accuracy: Catches errors in data, statistics, and citations, preventing costly misunderstandings.

Structural Cohesion: Ensures that headings, visuals, and executive summaries actually work together to explain your core message to people skimming the document .

Targeted Focus: Eliminates unnecessary information (throat-clearing) and reinforces the parts of the report that matter most to your audience

Activity	Questions Description	Mark
7	**List advantages and disadvantages of Word processing / desktop publishing**	8

Advantages:

*Ease of Editing: Simple to add, delete, or move text without retyping.

*Automated Corrections: Built-in spell check and grammar tools catch errors instantly.

*Templates: Readily available pre-designed templates make getting started quick and easy .

*Collaboration: Many word processors (like Google Docs) allow multiple users to edit in real-time.

*Disadvantages:

*Limited Graphics: Not built for complex, free-form image manipulation or multi-page layout designs.

Variable Formatting: Moving images or changing margins can disrupt the entire document flow.

Desktop Publishing (DTP

Advantages:

*WYSIWYG Layout: What you see on screen is precisely what prints out . Text automatically flows around and between frames, columns, and pages.

High-Level Design Control: DTP tools excel at handling intricate graphical elements, typography, and precise placement

Professional Output: Tools are postscript compatible and highly suited for commercial production of books, magazines, and catalogs.

Disadvantages:

Steep Learning Curve: Unlike basic Word Processors, dedicated DTP applications require substantial study to master all the design and layout tools .

High Cost: Professional software often requires expensive licenses, making them less accessible for casual use.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				

S.Picane 08 July 2026

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

LEARNER WORKBOOK

SAQA ID 13925

PRESENT INFORMATION IN A PUBLIC SETTING

Learner Information:

Details	Please Complete this Section
Name & Surname:	Simnikiwe Picane
Organisation:	PAICTA NSF
Unit/Dept:	
Facilitator Name:	Nomanesi Dinini
Date Started:	01 April 2026
Date of Completion:	08 July 2026

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OVERVIEW			
About the Learner Workbook	This Learner Exercise Workbook has been designed and developed to evaluate learners' level of understanding of the **PRESENT INFORMATION IN A PUBLIC SETTING** module. It forms part of a series of Learner Workbooks that have been developed for **National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS**		
Purpose	A person assessed as competent against this unit standard will be able to: Produce and respond to accessible written and oral communication in the workplace. This competence will equip individuals with the communication skills to operate more effectively in their workplace. It will contribute to more effective communication within the workplace environment.		
US No	**US Title**	**Level**	**Credits**
13925	**Present information in a public setting**	**5**	**5**
Context	This assessment represents the Formative Assessment component of the **Present information in a public setting** module, for the **National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS,** and should be completed in the classroom/training room.		
Resources	The following are resources needed for this assessment: 3. Learner Guide; and 4. Assessment Preparation.		
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the workbook to each learner; and • Interview the learner on similar questions, should he/she not be able to write.		

Instructions to Learners	Learners will be required to: • Complete the workbook as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame of.... hours.
Total Mark	This formative assessment carries a total mark of _ **points**. In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.
Equipment	Learners are required to have the following equipment in order to complete this workbook: • Pen and Pencil; • Ruler; and • Exam Pad – for additional paper.

3. GENERAL INFORMATION

LEARNER DETAILS		
Learner Full Names	Simnikiwe Picane	
Learner ID No.:	0106181295085	
Organisation:	PAICTA NSF	
Unit/Dept:		
Contact Details:	**Telephone /Cell Numbers:**	**Email Address:**
	0710247715	Simnikiwepicane06@gmail.com
WORKSHOP DETAILS		
Workshop Venue:		
Facilitator Name:		
Date Started:		
Date Completed:		

4. ASSESSMENT PREPARATION CHECKLIST

DESCRIPTION	YE	N	COMMENTS/
This assessment is a formative assessment and it is based on the outlined unit standard/s for the **Present information in a public setting** module.			
Your assessment evidence for the **Present information in a public setting** module needs to be submitted on..... (day) of.....(month).....(year) at the following address/			
You will be assessed based on the outlined Unit Standards. The assessment activities are linked to specific outcomes/assessment criteria of the outlined			
To determine your competence level, the following are the methods to be used for this assessment: 3. ..... 4.			
To be declared competent on **Present information in a public setting** module (formative assessment), you should have obtained at least 80% of the total mark			
You will be provided with detailed feedback on your performance of this assessment as follows: 3. Written Feedback 4. Verbal Feedback			
Should you be declared "not yet competent" on this assessment, you will be entitled for re-assessment opportunity/ies.			
You will be required to re-submit evidence (only for areas) you were declared not yet competent. A date for re-submission will be agreed with the			
You will be entitled to lodge an appeal should you not be satisfied with the assessment decision of your			

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Your results of assessment and portfolio of evidence information will not be provided to any person without your written consent.			

Learner's Declaration			
I…Simnikiwe Picane……herewith declare that I am ready for the assessment, that we have reviewed the assessment preparation and plan, I understand the assessment process and I am happy that the assessment will be			
Learner Signature:	**Date:**	**Facilitator Signature:**	**Date:**
S.Picane	08 July 2026		

Learning Unit 1 Present information in a public setting

UNIT STANDARD NUMBER	:	13925
LEVEL ON THE NQF	:	5
CREDITS	:	5
FIELD	:	Culture and Arts
SUB FIELD	:	Sport

PURPOSE:	Confidently conduct a presentation in various settings ranging from a corporate wellness program to a lecture, an informal discussion or a sports team.
LEARNING ASSUMED TO BE IN PLACE:	
Communication equivalent to NQF 4.	

SESSION 1.

Research and plan the content of the presentation in relation to the target audience.

Specific Outcomes

- Understanding of a presentation is demonstrated
- Researching and analyzing the target audience

Activity	Questions Description	Mark
1	What do you understand about presentations	5

The Core Elements

Every successful presentation hinges on five key components:

*Audience: The most critical factor. Understanding who you are speaking to and tailoring your message to their needs is essential for success.

*Purpose: A clear, crystal-straight objective. You should know exactly what you want the audience to learn or do by the end of your talk.

*Content: A strong, logical storyline that balances facts with relatable storytelling.

*Visuals: Aids like Microsoft PowerPoint or Google Slides that help clarify data, highlight key points, and keep audiences visually engaged.

Activity	Questions Description	Mark
2	List the three types of presentations	3

Delivery: Confident body language, proper pacing, strong vocal projection, and consistent eye contact.

Best Practices for Presenting

*Keep slides simple: Avoid overloading the screen with too much text or data. Use images and charts effectively.

Activity	Questions Description	Mark
3	Classify audiences into four categories	4

1.Demographic Audiences: Based on statistical characteristics such as age, gender, income level, and education level.

2.Psychographic Audiences: Classified based on psychological attributes like interests, values, lifestyle choices, and personality traits.

3.Behavioral Audiences: Categorized based on their behaviors and interactions, such as purchase behavior, usage rate, and brand loyalty.

4.Geographic Audiences: Classified based on physical location, including region, climate, language, and cultural background.
1

These categories help in tailoring communication and marketing strategies to effectively reach and engage different audience segments.

Activity	Questions Description	Mark
4	In pairs show comprehensive understanding of how to analyse your target audience	18

Pair 1 Demographics + Psychographics

*Demographics: Quantitative characteristics include age, gender, income, education, occupation, marital status, and location. These factors define the “who” of your audience in concrete terms.

*Psychographics: Qualitative traits such as interests, values, lifestyle, opinions, and attitudes reveal the “why” behind consumer behavior.

*Example: Targeting urban Gen Z women (demographics) who prioritize eco-friendly products (psychographics) enables precise messaging for sustainable fashion campaigns.

pair2 Customer Pain Points + Buyer Personas

*Customer Pain Points: Specific challenges or problems that potential customers face, e.g., lack of access, high cost, or insufficient guidance.

*Buyer Personas: Semi-fictional profiles based on collected demographic and psychographic data, illustrating motivations, goals, and challenges.

*Example: For a fitness app, Pain Point: low motivation, Persona: “Self-conscious young professional tracking workouts via mobile.”

pair 3 Behavioral Patterns + Purchase Drivers

*Behavioral Patterns: Actions like website navigation, social media engagement, buying frequency, and platform preferences.

*Purchase Drivers: Factors influencing buying decisions, including pricing, product features, convenience, social influence, and brand reputation.

Example: Users frequently abandoning online carts without discounts (behavior) may convert with targeted limited-time offers (driver).

pairs4 Example: Users frequently abandoning online carts without discounts (behavior) may convert with targeted limited-time offers (driver).

*Example: Users frequently abandoning online carts without discounts (behavior) may convert with targeted limited-time offers (driver).

Activity	Questions Description	Mark
5	Explain the characteristics to look out for from an audience	8

When analyzing an audience, it is crucial to consider their demographic characteristics and internal psychological traits. Demographic characteristics describe the outward features of the audiences, such as age, socio-economic level, profession, and more. These characteristics can influence how the age, socio-economic level, profession, and more. These characteristics can influence how the audience perceives and responds to the message. For example, the age of a group may affect their investment decisions, while socio-economic level may influence their financial priorities.

Psychological traits, on the other hand, refer to the audience's internal motivations and beliefs.

These can include their values, attitudes, and beliefs, which are essential for understanding how they will interpret and react to the presentation.

It is important to avoid stereotyping and totalizing, as these can be harmful to the relationship and ineffective as means of communication. Instead, focus on the unique features of the audience that can be leveraged to achieve the desired outcomes

39. SESSION 2.

40. PREPARE AND ORGANIZE PRESENTATION MATERIAL.

41. SPECIFIC OUTCOMES

- Preparing and organize the material to be used in the presentation is demonstrated
- Characteristics of a poor and unorganised presentation

Activity	Questions Description	Mark
6	In a group of three discuss no less than five points on how to prepare the material for presentation	**25**

1.Choose the right software for your needs

The easiest way to select the right presentation software for you is to simply find the one that is native to your device. For example, if you have a Mac, use Apple Keynote, if you work on Windows, use PowerPoint. Google Slides is recommended if you're working with someone, as it makes collaboration very easy. Another software option is Prezi: a specialty tool called Prezi that creates a presentation using motion, zoom, and panning across one giant visual space.

2. Organize your files

As you develop your script and visuals, you will need to start assembling all the assets for your slides. Create a unique folder on your computer to hold these items. Keep the folder organized by media type (presentation drafts, photos, videos, scripts) and back them up frequently to the Cloud or external disk. Label each file with a specific descriptive name, e.g. "Susan Johnson singing magpie 2020", as opposed to "IMG_4043.jpg", which can make it confusing to find your assets. The more organized you are up front, the easier preparing for your presentation will be.

3. Prepare your presentation materials

Make sure your presentation materials (script, graphics, actual slides) are saved in at least two safe spots (for example, your computer and an external USB drive) and are backed-up frequently. If you are using an online presentation software, such as Google Slides, be sure to also download a copy of your presentation in case the internet connection is unreliable. Having all the individual assets on hand in addition to your presentation slides can be helpful if you experience tech issues before presenting, or if you need to make any last minute changes. Make sure to label your final presentation with the title and your name so it's easy to find.

4. Practice, practice, practice!
Remember, practice makes perfect. People often run out of time making their presentations and have no time to practice. Most TED speakers practice at least ten times. Neuroscientist Jill-Bolte Taylor gave one of the most successful Talks in TED history with nearly 27 million views. How did she do it? She practiced her Talk over 40 times! By rehearsing multiple times you will naturally memorize your Talk, which means you won't need note cards when you give your final presentation.

5. Do a final test run
Before presenting, make sure the equipment you need is working properly. It's generally good practice to rehearse standing on the exact stage with the exact lighting using the exact computer that you will be using in your final presentation.

Best Practices
5 essential preparation steps for a successful presentation
by Tom Rielly • June 15, 2020

Keeping your presentation visuals minimalistic, simple, and clear is just one important step to remember when designing a hit presentation. Leaving nothing to chance, great presenters prove quite methodical as they prepare. Here's a checklist for everything you need to keep in mind before your next presentation:

1. Choose the right software for your needs
visualpres blogpost 2 softwares
The four main presentation softwares

The easiest way to select the right presentation software for you is to simply find the one that is native to your device. For example, if you have a Mac, use Apple Keynote, if you work on Windows, use PowerPoint. Google Slides is recommended if you're working with someone, as it makes collaboration very easy. Another software option is Prezi: a specialty tool called Prezi that creates a presentation using motion, zoom, and panning across one giant visual space.

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5. Do a final test run
Before presenting, make sure the equipment you need is working properly. It's generally good practice to rehearse standing on the exact stage with the exact lighting using the exact computer that you will be using in your final presentation.

Here's a quick checklist of what to look for when testing your equipment:

If you're not using your own computer, the one provided might be slower and have trouble playing media. If you have videos or other media, make sure they play correctly
Test the projector to make sure it's HD

Make sure images are clear
Test the sound of any clips you use, as this is what goes wrong most frequently
If you're using a mic, test the volume





SESSION 3.

Apply technical presentation techniques with or without technical equipment.

Specific Outcomes

- Apply technical presentation techniques with technical equipment
- Apply technical presentation techniques without technical equipment
- How to become a better presenter

Activity	Questions Description	Mark
7	Outline how a technical presentation should be done	7

How to Make a Technical Presentation
Creating a technical presentation involves several key components to ensure clarity, engagement, and understanding. Here's a structured outline to guide you through the proce

1.Preparation Phase: Define your purpose and audience, research, and organize content logically.

2.Structure of the Presentation: Include a title slide, introduction, background information, main content, data and evidence, and key findings or solutions.

3.Structure of the Presentation: Include a title slide, introduction, background information, main content, data and evidence, and key findings or solutions.

4Presentation Skills: Improve your presentation skills, choose appropriate backgrounds and fonts, and use images, videos, and animation effectively.

By following this outline, you can create a technical presentation that is both informative and engaging for your audience. .

Activity	Questions Description	Mark
8	Why is it important for a presenter to know the available equipment to you before hand	3

Knowing the available equipment beforehand is crucial for a presenter to ensure a smooth and professional presentation. It helps in avoiding technical issues that can disrupt the flow of the presentation. By being aware of the equipment, the presenter can make informed decisions about how to present the content effectively, ensuring that the message is conveyed clearly and engagingly to the audience. Additionally, knowing the equipment allows the presenter to troubleshoot any potential problems quickly, maintaining the audience's attention and the presentation's momentum.
storytellingpresentations.com
+5

Activity	Questions Description	Mark
9	Discuss ways to become a better presenter	15

To become a better presenter, consider the following tips:
Know Your Topic: Understand the subject thoroughly to convey your message clearly and confidently.
2
Outline Your Presentation: Create a structured outline to stay on track and keep the audience focused on key ideas.
2
Practice Regularly: Rehearse your presentation multiple times to build confidence and improve your delivery.
2
Use Visual Aids Wisely: Incorporate clean visuals and minimal text to enhance your message without overwhelming the audience.
2
Engage Your Audience: Use eye contact, active listening, and a natural flow to connect with your audience.
2

By implementing these strategies, you can significantly enhance your presentation skills and engage your audience effectively



SESSION 4.

DISPLAY OPTIMAL PRESENTATION SKILLS.

SPECIFIC OUTCOMES

- Outline optimal presentation skills
- Demonstrating areas for training and development of presentation skills

Activity	Questions Description	Mark
10	list five (5) of essential presentation skills	5

Five Essential Presentation Skills The five essential presentation skills are: 1.The five essential presentation skills are:

Engaging Language: Using words and phrases that fit the context and connect emotionally with the audience. 2.Eye Contact: Connecting intentionally with individuals throughout the audience to create stronger engagement.

3.Body Language: Communicating confidence through good posture and controlled movement.
4.Enthusiasm and Honesty: Showing excitement about the topic and being transparent with the audience.
5.Ability to Keep Things Simple: Making complex topics understandable and relatable to the audience.

Activity	Questions Description	Mark
11	Explain how preparation of a presentation should be done	8

Effective presentation preparation involves understanding your audience, clarifying your key message, structuring your content, designing supportive visuals, and practicing thoroughly.
Understand Your Audience
Before creating your presentation, research your audience's knowledge level, interests, and expectations. Tailor your content to address their needs and concerns, ensuring your message resonates and engages them from the start

. Consider what they already know, what questions they might have, and what objections or challenges they may raise

.

Define Your Core Message
Identify the single most important takeaway for your audience. Summarize your presentation in one clear sentence to ensure focus and clarity.

This helps guide the structure of your content and keeps your delivery concise and purposeful

Structure Your Content
Organize your presentation with a logical flow: introduction, main points, and conclusion. Map out a story arc or emotional journey to maintain attention, including moments of interest, surprise, clarity, and actionable takeaways
. This approach makes your presentation memorable and easier to follow.

Design Supportive Visuals

Create slides or props that enhance your spoken words without overwhelming the audience.

Use visuals to communicate ideas clearly, avoid clutter, and reinforce key points.Choose the right format and design elements to support your narrative effectively.

Practice Throughly

Rehearse your presentation multiple times out loud to build confidence and manage anxiety.Practise in the actual space if possible, and perform a technical run-through to check equipment, software, and timing .Iterative practise helps refine your delivery and ensures smooth execution.

Activity	Questions Description	Mark
12	In your own words describe how you would deal with presentation nervousness	10

Dealing with presentation nervousness involves a few key strategies:

1. **Preparation**: Thoroughly preparing for the presentation can significantly reduce anxiety. Knowing your material well helps build confidence. Practice in front of a mirror, record yourself, or present to friends or family for feedback.

2. **Practice Relaxation Techniques**: Techniques such as deep breathing, meditation, or visualization can help calm nerves. Before your presentation, take a few moments to breathe deeply and visualize yourself speaking confidently.

3. **Focus on the Message**: Shift your focus from how you are feeling to the message you want to convey. Remember that the audience is there to learn from you, not to judge you.

4. **Engage with the Audience**: Try to make a connection with the audience. Making eye contact, asking questions, or encouraging participation can help you feel more at ease.

5. **Start Small**: If possible, begin with a smaller group or informal setting to build your confidence. Gradually increasing the size of your audience can help ease you into larger presentations

6. **Embrace Imperfection**: Accept that mistakes can happen, and they don't have to derail your presentation. If you stumble, take a breath, and keep going. Most audiences are understanding and forgiving.

7. **Positive Self-Talk**: Replace negative thoughts with positive affirmations about your abilities and the value of your presentation. Remind yourself that you are knowledgeable and capable.

Activity	Questions Description	Mark
13	Explain the important features that will aid you to work your audience	6

To effectively work with your audience, consider the following important features:

Audience Analysis: Understand who you are giving a presentation to, who it is for, and what unique characteristics make it easier or harder for them to agree with or act on your message. This involves conceptualizing what your audience is like now and linking it with what you want them to do after hearing your presentation.

Engagement Techniques: Use storytelling, relatable anecdotes, and visual images to engage your audience. This can help them connect with your message and make it more memorable.

Tailored Content: Tailor your content to the interests, expectations, and level of understanding of your listeners. This ensures your key messages are conveyed effectively and resonates with your audience.

Effective Presentation: Presenting information to your audience is a crucial skill. You need to know who they are, what they already know, and what they need to know. This helps you design and deliver engaging and informative presentations.

Engagement Techniques: Techniques such as storytelling, visuals, interactive elements, and audience-centric approaches can keep your audience engaged throughout your presentation.

Audience Analysis: Knowing your audience—their beliefs, attitudes, age, education level, job functions, language, and culture—is the single most important aspect of developing your speech strategy and execution plan.

SESSION 5.

Perform self-monitoring and adapt the presentation.

Specific Outcomes

- Adjusting speech rate and volume and complexity of the content.
- Observe the response from the audience and adjust if necessary.
- Evaluate performance through the use of feedback tools.

Activity	Questions Description	Mark
14	Discuss how speech, complexity and volume rate can be adjusted	12

Speech, Complexity, and Volume Rate

Adjusting speech, complexity, and volume rate is essential for effective communication. Here are some strategies to enhance your public speaking skills:

Pitch Variation: Modulate your pitch to convey emotions and emphasize key points. A wide pitch range can add depth and expressiveness to your speech.

Volume Adjustment: Speak loudly enough to be heard by everyone in the audience, and adjust volume to overcome ambient noise. Use volume strategically to emphasize important points.

Pace Control: Maintain a comfortable rate of speech, aiming for 133-188 words per minute. Speed up to convey excitement or urgency, or slow down to highlight subtle points.

Pause Techniques: Use strategic pauses to control pace and allow the audience to process information. Vocal pauses can provide moments of reflection and emphasize, making a deep connection with your audience.

By mastering these techniques, you can enhance your public speaking, making your message more impactful and engaging.

Activity	Questions Description	Mark
15	Give a list of the structure that all presentations should follow	5

All effective presentations share a core structure: Introduction, Body, and Conclusion, with supporting elements to engage and guide the audience.

1. Introduction

The introduction sets the tone and captures attention. Key elements include:

Hook: A story, statistic, question, or bold statement to grab attention and create curiosity

Topic Statement: Clearly explain what the presentation is about

.

Purpose: State what the audience will learn, understand, or decide

.

Agenda/Overview: Outline the main points to help the audience follow the presentation

Activity	Questions Description	Mark
16	Explain why feedback is important	2

Feedback is essential because it enhances self-awareness, improves performance, strengthens relationships, and fosters a culture of continuous learning and accountability.

Enhances Self-Awareness

Feedback helps individuals identify their strengths and weaknesses, offering a clearer perspective on their abilities and behaviors. Since people are often emotionally attached to their work or habits, they may not objectively assess their performance. Receiving feedback allows for a more accurate understanding of one's skills and areas for improvement, which can boost confidence and motivation

Activity	Questions Description	Mark
17	Discuss ways how feedback should be delivered	6

Clarity and Specificity: Feedback is most effective when it is precise and relates to specific tasks or behaviors. Avoid vague statements; instead, provide concrete examples of what was done well or what needs improvement, so the recipient understands exactly what actions to continue or change

Timeliness: Deliver feedback promptly after the observed behavior or event. Immediate feedback ensures relevance and allows the recipient to act on it while the context is fresh

Benefits of Effective Feedback

When delivered effectively, feedback can increase motivation, improve performance, encourage continuous learning, strengthen relationships, and enhance personal growth. Employees are more likely to internalize constructive criticism and act on it when feedback is clear, specific, and supportive

.

By following these principles and techniques, feedback becomes a tool for development rather than a source of stress, fostering a culture of continuous improvement and accountability.

SESSION 6.

Determine and measure the outcome of the presentation.

Specific Outcomes

- Determine and measure the outcome of the presentation.

Activity	Questions Description	Mark
18	Explain how to measure presentation success	10

The success of a presentation is measured by whether it achieves its intended outcomes, engages the audience, and drives understanding or action.

Define Clear Objectives

Before presenting, establish specific, measurable goals. Success is not about applause or compliments but about achieving a defined outcome, such as persuading stakeholders, educating a team, or inspiring action . Use SMART goals: Specific, Measurable, Achievable, Relevant, and Time-bound. For example, “Increase audience understanding of the new product by 30%” or “Secure approval from three key decision-makers by the end of the week”

Measure Action and Behavioral Change

The ultimate measure of success is whether the presentation motivates action or changes behavior. This could include implementing new strategies, adopting recommended practices, or completing follow-up tasks. Tracking these outcomes over time provides evidence of long-term impact.

ASSESSOR REPORT		
UNIT STANDARD ID:	**UNIT STANDARD TITLE**	**CREDITS**
CANDIDATE NAME: DATE OF FEEDBACK:		
OVERALL ASSESSMENT DECISION: I _____, the assessor, declare the candidate **45.COMPETENT / NOT YET COMPETENT (CIRCLE RELEVANT) ON ALL THE CRITERIA WITHIN THE ASSIGNMENT.**		
STRENGTHS:		
WEAKNESSES:		

LEARNER COMMENTS:
DEVELOPMENT PLAN:
CANDIDATE DECLARATION: I _____, the candidate, declare that I have received feedback and been informed of my overall competence for the criteria within the assignment.
ASSESSOR SIGNATURE LEARNER SIGNATURE _____ _____

WEEK 3

National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

LEARNER WORKBOOK

SAQA ID 114051

Conduct a technical practitioners meeting

Unit Std #	114051
NQF Level	4
Notional hours	30
Credit(s)	3
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1 : Demonstrate knowledge of different types of technical practitioners meetings.
• 1. The demonstration describes the types of technical meetings and their uses. • 2. The demonstration identifies leadership styles used in meeting procedures. • 3. The demonstration identifies decision making processes used in meetings. • 4. The demonstration describes meeting conventions relevant to the type of meeting chosen. • 5. The demonstration identifies that the note taker should have technical background knowledge.
Specific Outcome 2 : Prepare for a technical practitioners meeting
• 1. The preparation ensures physical arrangements for the meeting is done, relevant to the type of meeting. • 2. The preparation ensures that meeting outcomes are clear, concise and well documented. • 3. The preparation ensures that meeting invitations are extended to relevant participants timeously. • 4. The preparation completes and distribute the meeting agenda and other supporting documentation needed for the type of meeting.
Specific Outcome 3 : Chair a technical practitioners meeting.
• 1. The chair and members agree on rules and guidelines on behavior. • 2. The chair applies agreed meeting conventions throughout the meeting, according to the type of meeting and in accordance with the standing procedures of the organisation(s) involved. • 3. The chair ensures that the published agenda is followed. • 4. The chair provides for active participation by all members to avoid/ minimize conflict. • 5. The chair ensures that meeting topics are prioritised and that discussion times are allocated according to importance, urgency, complexity and agenda. • 6. The chair ensures that agreed decisions are clear, accurate, includes a time frame for action and are within the mandate of the type of meeting conducted
• **Specific Outcome 4 : Conduct post meeting follow up for a technical meeting.**

- 1. The follow up ensures minutes of the meeting is produced accurately and in line with the policy of the organisation.
- 2. The follow up communicates agreed records of discussion to interested parties in a format and time frame that meet requirements of the type of meeting and of the organisation(s) involved.

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (3) x 10 = 30	9HRS	21HRS

1.	**Learning Programme Name:**	REFER TO COVER PAGE									
2.	**SAQA Qualification/ Unit Standard Title:**	REFER TO COVER PAGE									
3.	**Qualification/ Unit Standard**		**4.**	**SAQA ID Number**		**5.**	**N Q F Level**	5	**6.**	**Credits**	**3**
7.	**PURPOSE for offering this programme to your learners:**	REFER TO NEXT PAGE									
8.	**TARGET AUDIENCE for this specific programme:**	REFER TO NEXT PAGE									
9.	**Entry/Admission Requirements:**	REFER TO NEXT PAGE									

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 9hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =21 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				

Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				
4				
5				

Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

2				
3				
4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: Demonstrate knowledge of different types of technical practitioners meetings.
ASSESEMENT CRITERIA
❖ 1. The demonstration describes the types of technical meetings and their uses. ❖ 2. The demonstration identifies leadership styles used in meeting procedures. ❖ 3. The demonstration identifies decision making processes used in meetings. ❖ 4. The demonstration describes meeting conventions relevant to the type of meeting chosen. ❖ 5. The demonstration identifies that the note taker should have technical background knowledge.

Activity		SO	AC
1	**1.1In groups identify leadership styles used in meeting procedures: (5)** **1.2 In groups discuss why the note taker in a technical meeting should have technical background knowledge. (4)**	1	1-5

1.Autocratic (Authoritative) style

*Meeting Dynamics:Information flows one way (top-down). The leader speaks most of the time,sets rigid time limits, and makes final decisions without seeking consensus.

*Procedural Example: We have 10 minutes for this update. I have already decided we will launch the pilot next Monday. Let's move to the next item."

2. Democratic (Participative) style

*Procedural example: "We have two potential vendors.Let's go aroud the table to hear everyone's thoughts before we vote on how to proceed.

*Best For: Strategic planning, creative brainstorming, and solving complex problems that require team buy-in.

Technical background knowledge

*The Risk: A non-technical note-taker might misspell terms or confuse similar-sounding words (e.g., mistaking "SQL" for "sequel" in a specific context, or confusing "java" with "javaScript").

2.Distinguishing Signal from Noise

*The Risk: The notes become a transcript of the debate rather than a summary of the outcome.

3.Capturing implicit Context

*The Risk: The recorded action items may lack the necessary context, making them imposible to executive later.

4.Recognising Critical Action Items

*The Risk: Crucial system fixes, security patches, or refactoring tasks get lost because they weren't explicity stated as formal action items.

*The Technical Advantage: The note-taker recognises the impact of a bug or system dependency. They immediately flag it as a high-priority action item with an owner and a deadline.

SPECIFIC OUTCOME 2 :

Prepare for a technical practitioners meeting.

ASSESEMENT CRITERIA

- 1. The preparation ensures physical arrangements for the meeting is done, relevant to the type of meeting.
- 2. The preparation ensures that meeting outcomes are clear, concise and well documented.
- 3. The preparation ensures that meeting invitations are extended to relevant participants timeously.
- 4. The preparation completes and distribute the meeting agenda and other supporting documentation needed for the type of meeting.

Activity		SO	AC
2	**2. In groups discuss how to prepare for a technical practitioners meeting (10).**	2	1-4

SPECIFIC OUTCOME 3:

Chair a technical practitioners meeting.

ASSESEMENT CRITERIA

- 1. The chair and members agree on rules and guidelines on behavior.
- 2. The chair applies agreed meeting conventions throughout the meeting, according to the type of meeting and in accordance with the standing procedures of the organisation(s) involved.
- 3. The chair ensures that the published agenda is followed.
- 4. The chair provides for active participation by all members to avoid/ minimize conflict.
- 5. The chair ensures that meeting topics are prioritised and that discussion times are allocated according to importance, urgency, complexity and agenda.
- 6. The chair ensures that agreed decisions are clear, accurate, includes a time frame for action and are within the mandate of the type of meeting conducted.

Activity	Question	SO	AC
3	**3. In groups discuss how to chair a technical practitioners meeting (10)**	3	1-6



SPECIFIC OUTCOME 4:

Conduct post meeting follow up for a technical meeting.

ASSESEMENT CRITERIA

- 1. The follow up ensures minutes of the meeting are produced accurately and in line with the policy of the organisation.
- 2. The follow up communicates agreed records of discussion to interested parties in a format and time frame that meet requirements of the type of meeting and of the organisation(s) involved.

Activity	Question	SO	AC
4	**1. In pairs discuss how to conduct post meeting follow up for a technical meeting (10)**	4	1-2

Summarise Key Points

Document the main topics and decisions from the meeting, including the type of meeting, organization, date, time, facilitator, and adjournment time. Include technical details such as specifications, data, or diagrams shared during the discussion. This ensures the minutes serve as an official record of what was decided, not just what was said, providing clarity for all participants and stakeholders who were not present

2. List Action Items

Clearly outline tasks that need to be completed, specifying who is responsible and deadlines. Each action item should be explicit to avoid ambiguity, and any implied tasks should be formally captured during the meeting. This step bridges the gap between discussion and implementation, creating accountability and visibility

Distribute Meeting Minutes

Prepare a concise document that includes the summary, action items, and any relevant attachments or links. Send it to all participants and relevant stakeholders promptly. Use a structured format with categories like "Decisions," "Tasks," "Open Points," and "Next Steps" to make the document searchable and easy to reference

4. Encourage Feedback

Invite participants to provide feedback on the meeting and the follow-up document. This can be done via a short survey or email replies. Feedback helps improve future meetings and ensures that any misunderstandings are addressed early

Track Progress

Implement a system to monitor the completion of action items. Tools like Trello, Jira, OneNote, or Confluence can help track tasks, schedule reminders, and provide status updates. Regular check-ins, whether via email or brief status meetings, ensure that agreed measures are not forgotten and that the team maintains momentum

6. Maintain a Single Source of Truth

Ensure that all documentation is centralized and accessible to the team. Well-crafted recaps reduce confusion, prevent duplicated efforts, and provide context for those who could not attend. Striking a balance between comprehensive and concise documentation is key to driving action effectively

By following these steps, technical meetings can translate into actionable outcomes, improve accountability, and enhance team efficiency, ensuring that discussions lead to tangible results

7.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, ...declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

LEARNER WORKBOOK

SAQA ID 115431

Analyse feedback contexts and apply constructive feedback techniques

Unit Std #	115431
NQF Level	5
Notional hours	30
Credit(s)	3
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1 : Provide constructive feedback.
• The provision is selected in terms of type and channel • The provision ensures that the language, content and tone of feedback fit the situation, the occasion, the subject matter and the audience • The provision describes the behavior or event which is the focus of feedback, in neutral terms, with the absence of reference to character or personality of the recipient The provision identifies legal and ethical requirements that impact on the time and place where feedback is provided • The provision ensures that the content of the message meets the objectives set for providing feedback • The provision ensures that constructive feedback incorporates listening, open questions, paraphrasing and responding to receiver reactions
Specific Outcome 2 : Analyse feedback contexts and/ or situations
• The analysis identifies issues related to providing feedback in terms of context/ situation • The analysis identifies the purpose and/ or outcomes for specific feedback contexts/ situations • The analysis identifies consequences and/ or anticipated reactions for specific feedback contexts/ situations • The analysis ensures that the content and tone of feedback fit the situation, the occasion, the subject matter and the audience • The analysis identifies the legal and organisational requirements that impact on the type and manner of feedback provided
Specific Outcome 3: Respond constructively to feedback.

- The response ensures that the medium, language, content and tone fits the situation, the occasion, the matter and the audience
- The response clarifies feedback received, in terms of the objectives and issues raised
- The response evaluates feedback received, in terms of its applicability to the objectives and issues raised
- The response incorporates identifying options and/ or strategies for further action
- The response evaluated the outcome of feedback to determine if the objectives are met, or require further action

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (3) x 10 = 30	9HRS	21HRS

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7.	**PURPOSE for offering this programme to your learners:**	REFER TO NEXT PAGE									
8.	**TARGET AUDIENCE for this specific programme:**	REFER TO NEXT PAGE									
9.	**Entry/Admission Requirements:**	REFER TO NEXT PAGE									

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 9 hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =21 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

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- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
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5				

Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				

4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: PROVIDE CONSTRUCTIVE FEEDBACK.
ASSESEMENT CRITERIA
❖ The provision is selected in terms of type and channel ❖ The provision ensures that the language, content and tone of feedback fit the situation, the occasion, the subject matter and the audience ❖ The provision describes the behavior or event which is the focus of feedback, in neutral terms, with the absence of reference to character or personality of the recipient ❖ The provision identifies legal and ethical requirements that impact on the time and place where feedback is provided ❖ The provision ensures that the content of the message meets the objectives set for providing feedback ❖ The provision ensures that constructive feedback incorporates listening, open questions, paraphrasing and responding to receiver reactions

Activity	Practical	SO	AC
1	**1.1 1.1 In pairs describe how to provide constructive feedback. (10)**	1	1-5

To provide constructive feedback in pairs, consider the following steps:
Actively listen to your partner's perspective without interrupting and show empathy towards their point of view.
Be specific about the aspects you want to address, providing clear and actionable feedback.
Use the STAR/AR model to structure your feedback, focusing on specific behaviors or outcomes.
Start with a positive comment to set a supportive tone and end with another positive note to encourage and motivate your partner.
Encourage your partner to ask questions or seek clarification on the feedback provided and collaborate on setting actionable goals for improvement.
Schedule a follow-up discussion to track progress and provide further support if needed.
By following these steps, you can provide constructive feedback that is clear, specific, and focused on helping your partner improve their performance and develop their skills.

SPECIFIC OUTCOME 2 :

Analyse feedback contexts and/ or situations.

ASSESEMENT CRITERIA

- The analysis identifies issues related to providing feedback in terms of context/ situation
- The analysis identifies the purpose and/ or outcomes for specific feedback contexts/ situations
- The analysis identifies consequences and/ or anticipated reactions for specific feedback contexts/ situations
- The analysis ensures that the content and tone of feedback fit the situation, the occasion, the subject matter and the audience
- The analysis identifies the legal and organisational requirements that impact on the type and manner of feedback provided

Activity	Practical	SO	AC
2	**3. 1.1 In groups discuss how you would analyse feedback contexts and/ or situations (10)**	2	1-5

*Define the Trigger and Objective : Identify the specific event or behavior prompting the the feedback, and clarify what the communication needs to achieve.

*Evaluate the Audience and Tone: Ensure the language, content, and tone fit the situation, subject matter, and recipient. Keep descriptions neutral, separating the event from the persons character.

*Cluster into Themes: Groups comments to discover patterns and identify the frequency or extensiveness of specific issues, rather than relying on isolated remarks.

*Check constraints: Identify and legal, ethical, or organisational requirements that dictate the appropriate time, place, and manner for delivering the feedback.

*Anticipate Reactions: Predict the recipient's potential emotional responses or consequences, allowing the group to adjust the approach for a constructive outcome.

*Collaborate on Actionable Steps: Divide broad remarks into specific, actionable points, consulting subject-matter experts if needed to ensure accuracy

SPECIFIC OUTCOME 3:

Respond constructively to feedback.

ASSESEMENT CRITERIA

- The response ensures that the medium, language, content and tone fits the situation, the occasion, the matter and the audience
- The response clarifies feedback received, in terms of the objectives and issues raised
- The response evaluates feedback received, in terms of its applicability to the objectives and issues raised
- The response incorporates identifying options and/ or strategies for further action
- The response evaluated the outcome of feedback to determine if the objectives are met, or require further action

Activity	Question	SO	AC
3	**4. 1.1 In pairs describe how to respond constructively to feedback using examples. (8) (2)**	3	1-5

*Provide (Patner A) : "Your data slide was unclear during the stakeholder meeting,which caused confusion about our timeline."

*Receiver (Partner B - clarifying): "Thank you for pointing that out. Could you share exactly which metric or graphic caused the confusion so i can understand?"

*Receiver (Partner B - processing & Action): "I appreciate the feedback. Can i take until tommorow morning to review the slide deck and draft a simplified summary for the team?"

*Follow-up (Partner B): As suggested by resources on indeed, check in later to build a growth- oriented environment: "Here is the revised slide. Let's touch base to ensure thjis hits the mark."

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

LEARNER WORKBOOK

SAQA ID 114055

Demonstrate an awareness of ethics and professionalism

for the computer industry in South Africa

Unit Std #	114055
NQF Level	5
Notional hours	30
Credit(s)	3
Field	Field 10 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Contact Details

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Course:	

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Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
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3				
4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



Specific Outcome 1 : Describe professionalism for the computer industry in South Africa.
• 1. The description identifies acceptable and unacceptable professional practices found in the computer industry. • 2. The description identifies known professional bodies in South Africa. • 3. A short description of each named professional body is provided.
Specific Outcome 2 : Describe the codes of practice for professionalism in the IT industry in South Africa.
• 1. The description identifies the codes of practice for the IT industry in South Africa. • 2. The description provides a brief explanation of the codes of practice identified.
Specific Outcome 3 : Describe the code of ethics in the computer industry in South Africa.
• 1. The description confirms that the computer industry supports equality of opportunity. • 2. The description confirms the understanding that the computer industry is against computer software piracy. • 3. The description identifies ways in which piracy is addressed in South Africa.

SPECIFIC OUTCOME 1: Describe professionalism for the computer industry in South
ASSESEMENT CRITERIA
❖ 1. The description identifies acceptable and unacceptable professional practices found in the computer industry. ❖ 2. The description identifies known professional bodies in South Africa. ❖ 3. A short description of each named professional body is provided.

Activity	Question	SO	AC
1	1. **Describe professionalism for the computer industry in South Africa** 2. **List Computer science PROFESSIONAL BODIES IN SOUTH AFRICA.**	1	1,2

Technical Competence and Continuous Learning: Professionals are expected to maintain a strong foundation in relevant technologies, tools, and industry trends. This includes staying updated through workshops, certifications, online courses, and professional development programs to ensure high-quality work in areas such as software development, cybersecurity, IT support, and systems analysis

Ethical Standards and Integrity: Upholding ethical conduct is central to professionalism. This includes honesty, transparency, accountability, and respecting confidentiality and data privacy. Professionals must avoid conflicts of interest and act with integrity in all interactions with clients, colleagues, and stakeholders

Effective Communication and Collaboration: Clear and respectful communication is essential, both for technical and non-technical audiences. Professionals should actively listen, provide concise explanations, and collaborate effectively in teams, contributing positively to collective goals and resolving conflicts constructively

Quality of Work and Customer Focus: Delivering high-quality products and services that meet industry standards and client requirements is crucial. Professionals should prioritize client satisfaction, provide effective solutions, and maintain a professional image in all interactions

Several organizations in South Africa promote professionalism in the IT sector:

*Computer Society of South Africa (CSSA): Provides a code of conduct emphasizing integrity, competence, confidentiality, and professional development

*IITPSA (Institute of Information Technology Professionals South Africa): Offers frameworks for certification, professional development, and ethical standards.

*ISACA South Africa: Focuses on IT governance, security, risk management, and compliance, providing guidance for ethical and professional conduct

*ISO/IEC 27001: While international, it is widely adopted in South Africa for information security management, ensuring ethical handling of data

Summary

Professionalism in South Africa's computer industry is a combination of technical expertise, ethical behavior, continuous learning, effective communication, teamwork, and adherence to professional standards. By embodying these qualities, IT professionals build trust, maintain credibility, and contribute to a positive and thriving industry environment

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• 1. The description identifies acceptable and unacceptable professional practices found in the computer industry.			
• 2. The description identifies known professional bodies in South Africa.			



SPECIFIC OUTCOME 2 : Describe the codes of practice for professionalism in the IT industry in South Africa.
ASSESEMENT CRITERIA
❖ 1. The description identifies the codes of practice for the IT industry in South Africa.

Activity	Question	SO	AC
2	**Describe the CODE of practice for professionalism in the IT industry in South AS APPLIED TO** **1.1.A CONSULTANT** **1.2 SALESPERSONS**	2	1-2

IT consultants in South Africa are expected to uphold professionalism, competence, integrity, confidentiality, accountability, and compliance with laws and industry standards.

Professionalism

A consultant must act with professionalism at all times, demonstrating honesty, respect, and ethical behavior in interactions with clients, colleagues, and stakeholders. This includes maintaining a positive attitude, treating others courteously, and adhering to organizational and industry standards

Competency

Consultants are required to possess the necessary skills and expertise to perform their duties effectively. This involves obtaining relevant certifications, staying updated with emerging technologies, and engaging in continuous professional development to ensure high-quality service delivery

Confidentiality

Maintaining client confidentiality is critical. Consultants must protect sensitive information, avoid unauthorized disclosure, and comply with data protection regulations. This ensures trust and legal compliance in all professional engagements

Conflict of Interest

Consultants should disclose any potential conflicts of interest that could affect impartiality. This includes personal or financial relationships that may influence recommendations or decisions, ensuring transparency and ethical decision-making

Accountability

Consultants are responsible for their actions and decisions, taking ownership of their work, meeting deadlines, and delivering on commitments. They should be open to feedback and willing to learn from mistakes, reinforcing reliability and professional credibility

Communication

Effective communication is essential. Consultants must clearly articulate ideas and recommendations, actively listen to client needs, and provide regular updates on project progress. Transparency and clarity in communication enhance client trust and project success

Compliance

Consultants must adhere to all relevant laws, regulations, and industry standards, including intellectual property rights, data protection laws, and organizational policies. Compliance ensures ethical practice and reduces legal and reputational risks

Role of Professional Bodies

Professional bodies such as the Computer Society of South Africa (CSSA) and the Institute of Information Technology Professionals South Africa (IITPSA) provide frameworks for ethical conduct, professional development, and certification. They offer guidance on codes of conduct, including integrity, competence, and accountability, and support consultants in resolving ethical dilemmas

Summary

For IT consultants in South Africa, the code of practice emphasizes integrity, competence, confidentiality, accountability, effective communication, and compliance. Adhering to these principles ensures professional credibility, client trust, and alignment with both legal and industry standards, while professional bodies provide oversight and guidance to maintain high standards in the IT sector

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• 1. The description identifies the codes of practice for the IT industry in South Africa.			
• 2. The description provides a brief explanation of the codes of practice identified.			



SPECIFIC OUTCOME 3: Describe the code of ethics in the computer industry in South Africa.
ASSESEMENT CRITERIA
❖ 1. The description confirms that the computer industry supports equality of opportunity. ❖ 2. The description confirms the understanding that the computer industry is against computer software piracy.

Activity	Question	SO	AC
3	**Demonstrate understanding of the following** **1.1 Confidentiality and Privacy** **1.2 Fraud and Misuse** **1.3 Patents and Copyright Law**	3	2

Confidentiality and Privacy

Confidentiality and privacy are two fundamental concepts in the realm of data protection. Confidentiality refers to the obligation to protect sensitive information from unauthorized access and disclosure. It is crucial in various fields, including healthcare, finance, and legal professions, where client trust and legal compliance are essential. Privacy, on the other hand, relates to an individual's right to control their personal information. It encompasses how data is collected, stored, used, and shared. Laws such as the General Data Protection Regulation (GDPR) in the European Union and the California Consumer Privacy Act (CCPA) in the United States provide guidelines regarding individual privacy rights. Organizations must implement privacy policies to ensure that individuals' rights are respected and upheld.

Fraud is the intentional deception, misrepresentation, or concealment of material facts to secure an unfair or unlawful gain, resulting in financial or legal harm to a victim. Misuse refers to the improper or unauthorized use of resources, assets, or information for purposes other than their intended use.

Patents

A patent is a legal protection granted to an inventor for a new and useful invention or process, giving the holder exclusive rights to make, use, sell, or license the invention for a set period, typically 20 years from the filing date

. Patents are designed to reward innovation and encourage inventors to disclose their creations to the public. To qualify for a patent, an invention must be:

*Novel: It must be new and not publicly disclosed before.

*Non-obvious: It should not be obvious to someone skilled in the relevant field.

*Useful: It must have a specific, practical utility

Patents can cover a wide range of inventions, including machines, devices, chemical compositions, and processes. For example, a new type of engine or a medical device could be patented to prevent unauthorized use or reproduction

Practical Implications

Understanding the distinction is crucial for creators, inventors, and businesses. Patents prevent others from exploiting an invention without permission, while copyrights prevent unauthorized copying or distribution of creative works

Both forms of protection help maximize the value of intellectual property, avoid infringement, and maintain control over creations. Consulting an intellectual property attorney is recommended to determine the appropriate protection for a specific work or invention

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• 1. The description confirms that the computer industry supports equality of opportunity.			
• 2. The description confirms the understanding that the computer industry is against computer software piracy.			
• 3. The description identifies ways in which piracy is addressed in South Africa.			

Look back at what you have learnt in this unit standard. Make any suggestions or comments that need to be considered.

Reflection No:	**1**		
Date started:		**Date completed:**	

11. Did this learning guide help you to develop or revive new skills?	Yes	No
12. Are the materials and training you received clear and easy to understand?	Yes	No
13. Did the mentor explain to you the purpose of this training?	Yes	No
14. Do you find your facilitator to be supportive, dedicated and motivated to develop you?	Yes	No
15. Do you enjoy the learning programme?	Yes	No
16. Are you still motivated to participate in this learning programme?	Yes	No
17. Is there sufficient time to cover your specific problem areas in detail?	Yes	No
18. Which information/material do you find most helpful/useful?		
19. Which specific skills and knowledge did you gain from this training?		
20. Do you have any suggestions or comments about the course that will help us to improve future training?		

I have improved

Very much	More than I expected	As I expected	Not as much as I expected	Not at all

I rate my overall experience of the learning guide as

Excellent	Good	Average	Below average	Poor

Thank you for your comments and valuable participation

Application for Assessment

First Attempt □ Second Attempt □ Third Attempt□ RPL □

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time		
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	Theory	Practical
Date of assessment:		
When would be the best time for the Assessment?		

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

What resources would you require to conduct the assessment? 4. Assessor toolkit 5. Theoretical assignment 6. Practical assessment
How would you sequence your activities? Explain 6. Record preparation process 7. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 8. Learner will write summative assessments 9. Conduct observation in workplace where applicable 10. Complete assessor toolkit

WEEK 4

National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

LEARNER GUIDE

SAQA ID 114059

Demonstrate an understanding of estimating a unit of work and the implications of late delivery

Unit Std #	114059
NQF Level	4
Notional hours	50
Credit(s)	5
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
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- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

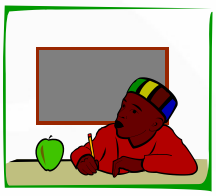
Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: **Demonstrate the ability to interpret given cost/benefit analysis documentation.**
ASSESEMENT CRITERIA
❖ 1. The demonstration identifies different parts of a cost/benefit analysis. ❖ 2. The demonstration explains the purpose of the different parts of a cost/ benefit analysis.

Activity	Practical	SO	AC
1	**1.1 In pairs demonstrate the ability to interpret given cost/benefit analysis documentation. (5)**	1	1-2

To effectively execute this as a team:Define the Scope:

Outline the project goals and decision context Cost benefit analysis: Steps and examples explained (2026).

Categorize Costs & Benefits: Partner 1 lists all Direct Costs (equipment, labor) and Indirect Costs (overhead) What Is Cost-Benefit Analysis? 4 Step Process. Partner 2 lists tangible revenue streams and intangible

benefits Cost-Benefit Analysis Explained: Usage, Advantages, and ....

Monetary Valuation: Assign a "common currency" (dollar/ZAR amount) to every line item What is a Cost-Benefit Analysis? Process Steps, Pros & Cons.

Calculate Metrics: Apply financial formulas to compute the Net Present Value (\(NPV\)) and the Benefit-Cost Ratio (\(BCR\)) Cost-Benefit Analysis: Step-by-Step Guide & Examples - Rippling.

Sensitivity Analysis & Recommendation: Test how altering assumptions (e.g., timeline shifts) impacts project viability before reaching a final verdict Cost-Benefit Analysis (CBA): Examples, Benefits and Steps | Indeed.com.

SPECIFIC OUTCOME 2 :

Prepare a time estimate for an element of work.

ASSESEMENT CRITERIA

- 1. The time estimate is based on a breakdown of the component in the logical parts for estimating.
- 2. The time estimate is based on an understanding that the estimate should include the implementation/testing of interfaces to other components where applicable.

Activity	Practical	SO	AC
2	**4. In pairs discuss how to estimate time accurately (10)** **5. Describe how to Prepare a work Schedule**	2	1-2

Break Down Tasks

Start by dividing larger tasks into smaller, manageable subtasks. Assess the complexity of each subtask, as more complex tasks may require additional planning or problem-solving time. This approach allows you to estimate each component more precisely and reduces the risk of underestimating the total duration

Use Historical Data

Reflect on past experiences with similar tasks to gauge how long they took. If working in a team, consult colleagues who have completed similar tasks to gather insights. Historical data provides a realistic baseline for your estimates and helps identify patterns in task duration

consider Potential Obstacles

Identify possible risks or interruptions that could delay progress. Adding buffer time to your estimates ensures that unexpected challenges do not derail your schedule. This is especially important for tasks that are unpredictable or involve multiple dependencies

Apply Estimation Techniques

Several techniques can improve accuracy:

Triangulation: Estimate the best-case, worst-case, and most likely scenario to create a range of expected completion time.

Three-Point Estimation: Use the formula (Optimistic + 4 × Most Likely + Pessimistic) ÷ 6 to calculate a realistic estimate.

Planning Poker: In team settings, engage everyone in estimating tasks to incorporate diverse perspectives

Time Tracking and Fudge Ratios: Track how long tasks actually take and adjust future estimates using a correction factor based on past under- or overestimation.

Use Mental Strategies
For quick or daily time estimates:

Rounding: Round times to the nearest 5, 10, or 15 minutes to simplify calculations.

Reference Points: Use familiar time markers like "half past" or "quarter to" as anchors.

Chunking: Break activities into smaller, known segments and sum their durations

Analogies: Compare tasks to familiar routines, e.g., "a TV commercial break is about 2 minutes".

Review and Adjust
Regularly review your schedule and completed tasks to identify areas for improvement. Adjust future estimates based on lessons learned, and remain flexible to accommodate unexpected changes.

By combining these strategies—task breakdown, historical data, risk assessment, structured estimation techniques, and mental strategies—you can improve the accuracy of your time estimates and manage tasks more effectively.

A work schedule is prepared by identifying tasks, prioritizing them, allocating time or shifts, and using tools or templates to organize and track responsibilities efficiently.

Steps to Create a Work Schedule

1. Define the Purpose and Scope
Determine whether the schedule is for personal tasks or managing a team. Clarify your goals, such as completing a project by a deadline or balancing team workloads, and align tasks with priorities to ensure focus and efficiency.

2. List All Tasks and Responsibilities
Write down every task, including one-time and recurring activities. Break larger tasks into smaller, actionable steps. For team schedules, involve members to capture all responsibilities and deadlines.
3. Assess Availability and Constraints
For team schedules, know each employee's working hours, contract type, and any mandatory days off. For personal schedules, consider your peak productivity times and personal commitments.

4. Prioritize Tasks
Sort tasks by importance and urgency. Tools like the Eisenhower Matrix can help categorize tasks into urgent, important, or less critical, ensuring essential work is completed on time.

5. Allocate Time or Shifts
Assign tasks or shifts based on priority and availability. Start with full-time or longer shifts, then fill in part-time or flexible slots. Avoid back-to-back shifts without adequate breaks to prevent fatigue.

6. Choose a Format and Tools
Decide between digital or physical formats. Digital tools like Google Calendar, Google Sheets, Notion, Todoist, or Asana allow easy updates, collaboration, and tracking. Templates can simplify the process and provide a clear visual structure.

7. Review and Adjust
Check for conflicts, workload balance, and compliance with labor regulations. For teams, share the schedule for feedback and make adjustments as needed. Regularly update the schedule to reflect changes in tasks or availability.

8. Communicate Clearly
Ensure everyone involved understands their responsibilities, deadlines, and shifts. Transparency reduces confusion, prevents favoritism, and promotes a healthy work-life balance.

SPECIFIC OUTCOME 3:

Prepare a cost estimate for an element of work.

ASSESEMENT CRITERIA

- 1. The cost estimate is based on a breakdown of the component in the logical parts for estimating.
- 2. The cost estimate is based on an understanding that the estimate should include the implementation/testing of interfaces to other components where applicable.
- 3. The cost estimate demonstrates an understanding that being assisted by other resources or people could escalate the costs.

Activity	Question	SO	AC
3	**5. Identify and explain cost estimate to develop a software system (7)** **6. List factors affecting software pricing (5)**	3	1-3

Key Factors Influencing Cost Estimation

The cost of developing a software system depends on several factors:

*Project Size: Measured in lines of (LOC) or Function Points (FP), larger projects require more effort and time

*Complexity: High-complexity projects demand specialized expertise and careful planning,increasing costs.

*Risk and Uncertainty: Unclear requirements, unfamiliar technologies, or inexperienced teams an introduce risks, leading to higher costs.

*Development Duration: The time required for design, development, testing, and deployment directly affects costs.

*Team size and Skill Level: Larger teams or highly skilled professionals typically result in higher labor costs.

Licenses, usage constraints
feature entitlements, support commitments

Software Pricing Strategies

Software pricing is a framework that determines how much an organization pays to adopt, use, and expand software over a given period. It includes a blend of licenses, usage constraints, feature entitlements, support commitments, along with renewal mechanics that shift as needs expand. Understanding pricing helps demarcate long-term value, reveals how spending grows, where unforeseen costs emerge, and whether software can sustain enterprise demands without undermining budgets.

SPECIFIC OUTCOME 4:

Demonstrate an understanding of the effect of late delivery of an element of work.

ASSESEMENT CRITERIA

- 1. The demonstration confirms the understanding that an element delivered is often a subset of a bigger deliverable.
- 2. The demonstration explains the effect of late delivery on other related components.

Activity	Question	SO	AC
4	**2. In pairs demonstrate an understanding of the effect of late delivery of an element of work. (5)**	4	1-2

Late delivery of a work element can disrupt project timelines, increase costs, strain resources, and negatively affect team performance and stakeholder relationships.

Impact on Project Timeline

When an element of work is delivered late, it can create a domino effect, delaying subsequent tasks that depend on its completion. This is particularly critical in projects with highly interdependent tasks, as one delay can cascade through the entire project, potentially extending the overall timeline and causing missed deadlines.

In collaborative environments, such as group projects, late delivery can hinder the progress of other team members, affecting the readiness and effectiveness of the team

Increased Costs

Late delivery often leads to additional labor costs, as teams may need to work overtime to catch up

. It can also result in financial penalties if contractual obligations are not met, and may damage client relationships, potentially affecting future business opportunities

. Resource reallocation to address delays can further increase costs and reduce efficiency.

Resource Allocation and Team Dynamics

Delays may require reallocation of resources, impacting other areas of the project and creating inefficiencies.

Teams may experience stress and reduced morale as they adjust to new timelines, which can affect productivity and collaboration.

Negative group dynamics may arise if accountability is not clearly established.

Quality and Stakeholder Impact

Late delivery can compromise the quality of work, as rushed efforts to meet deadlines may lead to errors or incomplete outputs.

Stakeholder satisfaction may decline if expectations are not met, emphasizing the importance of proactive communication and progress monitoring.

Mitigation Strategies

To minimize the effects of late delivery, it is essential to:

Set clear deadlines and communicate expectations effectively

Monitor progress regularly and address issues promptly

Implement contingency plans and allocate buffer time for critical tasks.

Encourage accountability and address underlying causes of delays to improve collaboration

Understanding these effects helps teams manage timelines, costs, and resources more effectively, ensuring smoother project execution and better outcomes.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
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4				
5				
6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115391

Demonstrate an understanding of the principles of the internet and the world-wide-web

Unit Std #	115391
NQF Level	4
Notional hours	30
Credit(s)	3
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1 : Explain the principles of the internet and the world-wide-web.
• The explanation outlines the origins and history of the internet • The explanation identifies the major applications of the internet • The explanation demonstrates the use of major internet applications • The explanation describes the history and development of the world-wide-web
Specific Outcome 2 : Explain how the world-wide-web incorporates the various internet applications.
• The explanation provides a comprehensive understanding of the physical context of web pages • The explanation identifies how the world-wide-web can be applied in an intranet and extranet • The explanation describes the latest internet applications, including web-based email, instant messaging and Voice-over-IP (VoIP)

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (3) x 10 = 30	9HRS	21HRS

1.	**Learning Programme Name:**	REFER TO COVER PAGE									
2.	**SAQA Qualification/ Unit Standard Title:**	REFER TO COVER PAGE									
3.	**Qualification/ Unit Standard**		**4.**	**SAQA ID Number**		**5.**	**N Q F Level**	4	**6.**	**Credits**	**3**
7.	**PURPOSE for offering this programme to your learners:**	REFER TO NEXT PAGE									
8.	**TARGET AUDIENCE for this specific programme:**	REFER TO NEXT PAGE									
9.	**Entry/Admission Requirements:**	REFER TO NEXT PAGE									

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 9hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =21 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	

Candidate Details			
Name		**ID No**	
Job		**Company**	
Department			

Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
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Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

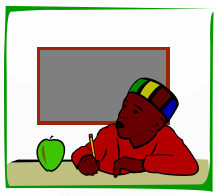
Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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2				
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4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: Explain the principles of the internet and the world-wide-web.
ASSESEMENT CRITERIA
❖ The explanation outlines the origins and history of the internet ❖ The explanation identifies the major applications of the internet ❖ The explanation demonstrates the use of major internet applications ❖ The explanation describes the history and development of the world-wide-web

Activity	Practical	SO	AC
1	**1.1 In pairs discuss the principles of the internet and the world-wide-web (10).**	1	1-4

When discussing these concepts in pairs, consider exploring their foundational layers and differences:Infrastructure vs. Content: The Internet is the global highway (cables, routers), whereas the Web is the vehicles, signs, and destinations (web pages, media) traveling on it. The Internet can exist independently (for emails, file sharing, VoIP), but the Web cannot function without the Internet's network foundation.

Decentralization & Openness: The Internet is inherently decentralized with no single governing body. The Web operates on universal access, allowing anyone to publish or retrieve information seamlessly through open standards and interoperability.

Addressing and Protocols: The Web relies on URIs/URLs to uniquely identify resources and HTTP to request and transfer hypertext documents.

Statelessness: Hypertext Transfer Protocol is stateless, meaning each web request is independent—servers drop the connection immediately after fulfilling a request unless cookies or sessions are applied.

SPECIFIC OUTCOME 2 : Explain how the world-wide-web incorporates the various internet applications.
ASSESEMENT CRITERIA
❖ The explanation provides a comprehensive understanding of the physical context of web pages ❖ The explanation identifies how the world-wide-web can be applied in an intranet and extranet ❖ The explanation describes the latest internet applications, including web-based email, instant messaging and Voice-over-IP (VoIP)

Activity	Question	SO	AC
2	**3. In pairs discuss how the world-wide-web incorporates the various internet applications. (10)**	2	1-3

The World Wide Web integrates internet applications in pairs by linking client-side tools like web browsers with server-side services, and combining web applications with APIs to create interactive, interconnected experiences.

Web Browsers and Web Servers

Web browsers (such as Chrome, Firefox, and Safari) act as the client-side interface, allowing users to access web content, while web servers store and deliver web pages and applications. This pairing enables users to request information and receive dynamic content, forming the backbone of web interaction and supporting applications like email, online banking, and social media platforms

Web Applications and APIs

Web applications, including Gmail, Google Docs, and online banking platforms, run in browsers and provide interactive services. These applications often pair with APIs (Application Programming Interfaces), which allow them to communicate with other services, retrieve data, or integrate third-party functionalities. For example, a travel booking site may use an API to access airline databases, combining two applications into a seamless user experience

Social Media and Multimedia Integration

Social media platforms like Facebook, Instagram, and Twitter pair with multimedia applications to deliver content such as videos, images, and live streams. This integration allows users to interact, share, and consume content in real time, demonstrating how the WWW combines communication tools with multimedia services

content Management Systems and Collaboration Tools

Content Management Systems (CMS) like WordPress or Drupal pair with collaborative web applications such as Google Docs or Microsoft Office Online. This combination enables users to create, edit, and publish content dynamically while supporting real-time collaboration and user interactivity.

Hyperlinks and Navigation

Hyperlinks connect web pages and applications, allowing users to navigate between different services seamlessly. This pairing of hypertext and web applications ensures that users can move from one application to another, such as from a news site to a social media platform, without leaving the web environment

Summary

The WWW functions as a platform that pairs client-side and server-side applications, web applications with APIs, and social media with multimedia tools, creating an interconnected ecosystem. These pairings enable dynamic content delivery, real-time interaction, and seamless navigation, making the web a versatile and integrated environment for diverse internet applications

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR

I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

WEEK 5

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115358

Apply information gathering techniques for computer system development

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	

Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	

Candidate Details				
Name			**ID No**	
Job			**Company**	
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				
4				
5				

Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				

4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



Specific Outcome 1 : Design and conduct an interview for gathering information for computer system development.
• The interviewee reports that he or she has understood the interview objectives. • The interviewee reports that he or she has understood the interview questions. • The interview provides answers that meet interview objectives. • The presentation of the interview is appropriate to the interviewee.
Specific Outcome 2 : Design and perform an analysis of the results from a questionnaire for gathering information.
• The respondents report that they understand the questionnaire objectives. • The respondents report that they understand the questions. • The questionnaire responses provide answers that meet questionnaire objectives. • The presentation of the questionnaire is appropriate to the target population. • A summary of questionnaire responses, and a comparison with expected responses, allows summary statements to be made about the population sample.
Specific Outcome 3 : Gather data from documents for computer system development.
• Research notes identify data that meet the specified information requirements using an industry recommended format. • Research notes identify the characteristics of the data and the relationships between data items. • Research notes identifying data items facilitate access to those data items.
Specific Outcome 4 : Observe a person's behaviour for gathering information for computer system development
• A record of the behaviour identifies events that meet the specified information requirements, and outlines those events. • A report about the observation compares the outcome of the observation with the observation objectives.
Specific Outcome 5 : Consolidate the information gathered via different techniques.

- The comparison identifies agreement and differences between the information gathered from different techniques.
- Differences are resolved and justified by reviewing the information gathering techniques.

SPECIFIC OUTCOME 1: Design and conduct an interview for gathering information for computer system development.
ASSESEMENT CRITERIA
❖ The interviewee reports that he or she has understood the interview objectives. ❖ The interviewee reports that he or she has understood the interview questions. ❖ The interview provides answers that meet interview objectives. ❖ The presentation of the interview is appropriate to the interviewee.

Activity	Practical	SO	AC
1	**1.1 Why should you conduct interviews? (2)** **1.2 Explain when interviews are not the best option (2)** **1.3 Design an interviewee report (10)**	1	1-4

1. Evaluate Skills, Experience, and Qualifications

Interviews help verify a candidate's work history, technical abilities, and soft skills through targeted questions.

They reveal problem-solving ability, critical thinking, and communication style that r

*They reveal problem-solving ability, critical thinking, and communication style that r

2. Assess Cultural and Organizational Fit

Interviews show whether a candidate aligns with company values, work style, and team dynamics.

3. Observe Non-Verbal Communication and Interpersonal Skills

Body language, confidence, attitude, and professionalism are visible only in conversation.

These traits help predict how candidates will interact with colleagues and customers.

4. Provide a Two-Way Evaluation

Candidates use interviews to understand the company's culture, expectations, and work environment.

This mutual assessment helps both sides determine fit and reduces turnover.

5.. Clarify Job Expectations and Gather Additional Information

Interviews allow employers to explain duties, expectations, and workplace norms clearly.

I

They also help uncover additional insights about candidates that may not appear in written materials.

6.Ensure Fair, Informed, and Legally Sound Hiring Decisions

Structured interviews support fairness, consistency, and reduced bias.

They help verify facts, confirm identity, and meet organizational hiring standards.

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• The interviewee reports that he or she has understood the interview objectives.			
• The interviewee reports that he or she has understood the interview questions.			

SPECIFIC OUTCOME 2 : Design and perform an analysis of the results from a questionnaire for gathering information.
ASSESEMENT CRITERIA
❖ The respondents report that they understand the questionnaire objectives. ❖ The respondents report that they understand the questions. ❖ The questionnaire responses provide answers that meet questionnaire objectives. ❖ The presentation of the questionnaire is appropriate to the target population. ❖ A summary of questionnaire responses, and a comparison with expected responses, allows summary statements to be made about the population sample.

Activity	Practical	SO	AC
2	**6. What are the Advantages and Disadvantages of a questionnaire (6)** **7. List Two types of questionnaires (2)** **8. Design a questionnaires with questionnaire responses that provide answers that meet questionnaire objectives (10)**	2	1-5

A well-designed questionnaire starts with clear objectives, structured questions, and response options that directly support the research goals.

Step 1: Define Research Objectives

Before drafting questions, clearly outline what you want to learn. For example, if your goal is to measure customer satisfaction, your objectives might include assessing service quality, responsiveness, and overall experience.

. Every question should map directly to one of these objectives to avoid collecting irrelevant data

advantages of questionnaires include:

Cost-effective: They are generally cheaper to administer than other data collection methods

Efficiency: Questionnaires can collect large amounts of data quickly from many respondents

Standardization: They provide a consistent format for all respondents, reducing interviewer bias

Flexibility: They can be distributed in various formats (online, paper) and can reach a wide audience

disadvantages of questionnaires include:

Limited depth: They may not capture complex issues or provide in-depth insights

Response bias: Respondents may misinterpret questions or provide socially desirable answers

Low response rates: Some questionnaires may suffer from low participation, affecting data quality A

Design challenges: Poorly designed questionnaires can lead to confusion and unreliable data

types of questioners

-the most common questions have yes or no

-multiple choice questions

-return skills 1 to 5

-short answers questions

-you compare expect vs actual results

Question: "How satisfied are you with your current role?"
Responses: Very dissatisfied, Dissatisfied, Neutral, Satisfied, Very satisfied
Question: "Do you feel your contributions are recognized?"
Responses: Never, Rarely, Sometimes, Often, Always
Question: "What changes would improve your work experience?"
Responses: Open-ended text
Question: "I feel motivated to achieve my goals at work."
Responses: Strongly disagree, Disagree, Neutral, Agree, Strongly agree
By aligning each question and response with the research objectives, the questionnaire ensures that collected data is meaningful, analyzable, and actionable

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• The respondents report that they understand the questionnaire objectives.			
• The respondents report that they understand the questions.			
• The questionnaire responses provide answers that meet questionnaire objectives.			



SPECIFIC OUTCOME 3:

Gather data from documents for computer system development.

ASSESEMENT CRITERIA

- ❖ Research notes identify data that meet the specified information requirements using an industry recommended format.
- ❖ Research notes identify the characteristics of the data and the relationships between data items.
- ❖ Research notes identifying data items facilitate access to those data items.

Activity	Question	SO	AC
3	**7. What is the importance of documents analyst (2)** **8. What is Coding data (2)** **9. What is the importance of Organizing data (2)**	3	1-3

*Data Accuracy: They verify and validate the information contained in documents, which is essential for maintaining data integrity.

*Compliance: They ensure that documents adhere to legal and regulatory standards, helping organizations avoid penalties and legal issues.

*Efficiency: By organizing and categorizing documents, they improve workflow efficiency, making it easier for teams to access necessary information.

*Risk Management: They identify potential risks associated with document handling and storage, implementing strategies to mitigate these risks.

*Quality Control: Analysts conduct quality checks on documents to ensure they meet organizational standards, which enhances overall quality.

Document analysts are essential for extracting, organizing, and interpreting information from various types of documents, enabling informed decision-making and knowledge creation across different domains.

Coding data is a systematic process used in research to organize, interpret, and analyze raw data. In qualitative research, coding involves assigning descriptive or interpretive labels to chunks of text, such as words, phrases, sentences, or paragraphs, to capture their essence or meaning

Improved Efficiency

Well-organized data allows for quick access and retrieval, saving valuable time when searching for specific information. Structured data reduces redundancies and unnecessary steps, streamlining workflows and automating routine tasks, which leads to significant time savings for individuals and organizations

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• Research notes identify data that meet the specified information requirements using an industry recommended format.			
• Research notes identify the characteristics of the data and the relationships between data items.			
• Research notes identifying data items facilitate access to those data items.			



SPECIFIC OUTCOME 4:

Observe a person's behaviour for gathering information for computer system development

ASSESEMENT CRITERIA

- A record of the behaviour identifies events that meet the specified information requirements, and outlines those events.
- A report about the observation compares the outcome of the observation with the observation objectives.

Activity	Question	SO	AC
4	**4. Define ABC data collection (2)** **5. Give an example of ABC data collection (10)** **6. Comment on the use of ABC data collection tool (2)**	4	1-2

ABC data collection is a method used in Applied Behavior Analysis (ABA) to observe and record the antecedents, behaviors, and consequences surrounding a specific behavior to understand its function and guide interventions.

Antecedent: No paper or toner in the printer.
Behavior: Student drops the papers into the trash.
Consequence: The task is avoided, and the student escapes the frustrating situation.
This example illustrates how ABC data helps identify the function of behaviors, such as escape or attention-seeking, and guides the development of replacement behaviors or interventions

The ABC data collection tool is commonly used in behavioral analysis to gather information about antecedents, behaviors, and consequences. It helps in identifying patterns and triggers for specific behaviors. Here are some key points to consider when using the ABC data collection tool:

It provides a structured method for recording behavioral data in a systematic and objective manner.
The tool helps in understanding the environmental factors that may influence behavior.
By analyzing the data collected, patterns and trends in behavior can be identified, leading to the development of effective behavior intervention strategies.
It is important to ensure consistency and accuracy in recording data to obtain reliable results.

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• A record of the behaviour identifies events that meet the specified information requirements, and outlines those events.			
• A report about the observation compares the outcome of the observation with the observation objectives.			
• A record of the behaviour identifies events that meet the specified information requirements, and outlines those events.			

SPECIFIC OUTCOME 5: Consolidate the information gathered via different techniques.
ASSESEMENT CRITERIA
❖ The comparison identifies agreement and differences between the information gathered from different techniques. ❖ Differences are resolved and justified by reviewing the information gathering techniques.

Activity	Question	SO	AC
5	**1. Which data collection technique to apply depends on a number of factors, List those factors (4)** **2. What do you understand by Pattern evaluation (2)** **3. What do you understand by Data mining (2)**	5	1-2

The choice of data collection technique to apply depends on several factors, including the research objectives, nature of data, resources available, population characteristics, ethical considerations, research environment, data quality, and the researcher's skills. These factors collectively influence the selection of the most suitable data collection method for a research study.

Cross-validation and bootstrapping are robust validation techniques. Cross-validation partitions data into folds to test model generalization, while bootstrapping resamples with replacement to estimate performance variability. Stratified K-Fold ensures class proportions remain consistent across folds, improving reliability in imbalanced datasets.

Clustering evaluation uses external indices (e.g., Rand Index, Jaccard coefficient) when ground truth is available, and internal indices (e.g., Dunn index) when it's not. Stability and efficiency are also considered to ensure consistent clustering results.

Advanced evaluation metrics like AUC-ROC, logarithmic loss, and Matthews Correlation Coefficient provide deeper insights into model discrimination and probabilistic accuracy. Calibration plots and partial dependence plots help assess prediction probability alignment and feature influence.

Visual inspection remains valuable for detecting anomalies or confirming intuitive patterns, especially in smaller datasets. Running time is also measured to ensure scalability for large datasets.

Example:

```
from sklearn.model_selection import cross_val_score
from sklearn.metrics import precision_score, recall_score, f1_score
scores = cross_val_score(model, X, y, cv=5, scoring='accuracy')
print("Cross-validated Accuracy:", scores.mean())
```

Copy

In practice, combining multiple metrics and validation techniques yields the most trustworthy evaluation of predictive patterns.

Data mining is the process of discovering patterns, correlations, and anomalies in large datasets using statistical methods, machine learning, and pattern recognition to generate actionable insights . It transforms raw data into meaningful knowledge that supports decision-making, forecasting, and optimization across industries.

It evolved from early statistical analysis in the 1950s to Knowledge Discovery in Databases (KDD) in the 1980s, and now integrates with big data platforms, AI, and cloud computing for real-time, large-scale analysis .

Core Process Steps A typical data mining workflow follows a structured approach:

Problem Definition – Identify the business question and objectives.

Data Preparation – Collect, clean, and transform data from multiple sources.

Exploration – Use statistics and visualization to understand patterns.

Model Building – Apply algorithms like classification, clustering, regression.

Validation – Test model accuracy and generalizability.

Implementation – Deploy models into production systems.

Evaluation – Measure impact and refine models .

Key Techniques

Classification – Assigns data to predefined categories (e.g., spam detection).

Regression – Predicts continuous values (e.g., sales forecasting).

Clustering – Groups similar data points without labels (e.g., customer segmentation).

Association Rule Mining – Finds relationships between variables (e.g., market basket analysis).

Anomaly Detection – Identifies unusual patterns (e.g., fraud detection) .

Applications

Retail & E-commerce – Personalized recommendations, demand forecasting.

Finance – Fraud detection, credit risk assessment.

Healthcare – Early disease detection, treatment optimization.

Manufacturing – Predictive maintenance, process optimization .

Self-check			
OUTCOME	**Yes**	**No**	**I Need help**
• The comparison identifies agreement and differences between the information gathered from different techniques.			
• Differences are resolved and justified by reviewing the information gathering techniques.			
• The comparison identifies agreement and differences between the information gathered from different techniques.			

Look back at what you have learnt in this unit standard. Make any suggestions or comments that need to be considered.

Reflection No:	**1**		
Date started:		**Date completed:**	

21. Did this learning guide help you to develop or revive new skills?	Yes	No
22. Are the materials and training you received clear and easy to understand?	Yes	No
23. Did the mentor explain to you the purpose of this training?	Yes	No
24. Do you find your facilitator to be supportive, dedicated and motivated to develop you?	Yes	No
25. Do you enjoy the learning programme?	Yes	No
26. Are you still motivated to participate in this learning programme?	Yes	No
27. Is there sufficient time to cover your specific problem areas in detail?	Yes	No
28. Which information/material do you find most helpful/useful?		
29. Which specific skills and knowledge did you gain from this training?		
30. Do you have any suggestions or comments about the course that will help us to improve future training?		

I have improved

Very much	More than I expected	As I expected	Not as much as I expected	Not at all

I rate my overall experience of the learning guide as

Excellent	Good	Average	Below average	Poor

Thank you for your comments and valuable participation

Application for Assessment

First Attempt □ Second Attempt □ Third Attempt□ RPL □

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time		
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	Theory	Practical
Date of assessment:		
When would be the best time for the Assessment?		

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

What resources would you require to conduct the assessment? 7. Assessor toolkit 8. Theoretical assignment 9. Practical assessment
How would you sequence your activities? Explain 11. Record preparation process 12. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 13. Learner will write summative assessments 14. Conduct observation in workplace where applicable 15. Complete assessor toolkit

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115365

Apply the principles of designing computer system inputs and outputs

Unit Std #	115365
NQF Level	5
Notional hours	70
Credit(s)	7
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1 : Explain the principles of computer input and output design.
• The explanation identifies the various types of inputs and outputs. • The explanation distinguishes between appearance and the underlying structure and processes. • The explanation identifies the purpose of user involvement in creating the designs. • The explanation compares online computer functions with manual forms and offline data entry. • The explanation includes a discussion of input and output that compares graphical input and output functions with text based input and output functions.
Specific Outcome 2 : Design computer input and output functions
❖ The design meets the specification for the function. ❖ The design can be implemented in the specified computer environment. • The design conforms to an industry recommended format for the function.
Specific Outcome 3 : Create computer input and output functions
• The creation ensures that the function format corresponds to the design. • The creation ensures that the function behaviour corresponds to the design.

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (7) x 10 = 70	6HRS	14HRS

1.	**Learning Programme Name:**	REFER TO COVER PAGE									
2.	**SAQA Qualification/ Unit Standard Title:**	REFER TO COVER PAGE									
3.	**Qualification/ Unit Standard**		**4.**	**SAQA ID Number**		**5.**	**N Q F Level**	4	**6.**	**Credits**	**2**
7.	**PURPOSE for offering this programme to your learners:**	REFER TO NEXT PAGE									
8.	**TARGET AUDIENCE for this specific programme:**	REFER TO NEXT PAGE									
9.	**Entry/Admission Requirements:**	REFER TO NEXT PAGE									

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 6hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =14 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				

Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

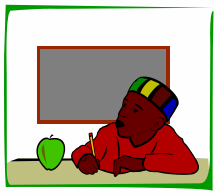
Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				
4				
5				

Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

2				
3				
4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: Explain the principles of computer input and output design.
ASSESEMENT CRITERIA
❖ The explanation identifies the various types of inputs and outputs. ❖ The explanation distinguishes between appearance and the underlying structure and processes. ❖ The explanation identifies the purpose of user involvement in creating the designs. ❖ The explanation compares online computer functions with manual forms and offline data entry. ❖ The explanation includes a discussion of input and output that compares graphical input and output functions with text based input and output functions.

Activity	Practical	SO	AC
1	**1.1 In pairs explain the principles of computer input and output design. (10)**	1	1-5

1. Clarity and Simplicity
Input: Input fields, buttons, and controls should be clearly labeled and simple, so users understand what is required without confusion. Logical organization and minimal complexity improve usability
Output: Output should present information clearly, using straightforward language and avoiding unnecessary technical jargon, ensuring users can easily interpret results

3.Consistency
Input: Maintain uniform terminology, layout, and interaction patterns across all input interfaces to help users predict system behavior.
Output: Keep output formats and styles consistent, allowing users to compare and understand information efficiently over time

4. Feedback
Input: Provide immediate feedback for user actions, such as validation messages, error alerts, or confirmation notifications, to guide correct data entry

Output: Clearly indicate the results of operations, using visual cues like progress bars or status messages to inform users about system processing.

4. Accessibility

Input: Design input controls that are accessible to all users, including those with disabilities, supporting screen readers and keyboard navigation

Output: Ensure output is available in multiple formats (visual, audio, or text) to accommodate diverse user needs

5.Error Prevention and Recovery

Input: Implement mechanisms to prevent errors, such as auto-complete, drop-down suggestions, and validation checks, and provide easy ways to correct mistakes

*Output: Present error messages or alerts clearly, guiding users to understand and resolve issues without confusion.

6. Relevance and Efficiency

Input: Collect only necessary data to reduce user effort and avoid overwhelming the system.

Output: Present concise, relevant information that supports decision-making, avoiding information overload and focusing on user tasks.

SPECIFIC OUTCOME 2 :

Design computer input and output functions.

ASSESEMENT CRITERIA

- The design meets the specification for the function.
- The design can be implemented in the specified computer environment.
- The design conforms to an industry recommended format for the function.

Activity	Practical	SO	AC
2	**9. Describe the following, Electromagnetic identification, Point-of-sale terminals, Smart card applications (6)** **10. List Types of Outputs (5)** **11. List System User Issues for Output Design (5)**	2	1-3

Key ConceptsElectromagnetic Identification (RFID/NFC): A reader generates an electromagnetic field, which powers the microchip inside an object or card without physical contact, allowing data exchange over short distances.

Point-of-Sale (POS) Terminals: Computer-based hardware and software systems (e.g., Yoco POS in South Africa) used by merchants to process card and digital wallet transactions, record sales, and update inventory.

Smart Card Applications: Physical cards embedded with an integrated chip. They are used globally for secure transactions, such as the South African Smart ID Card, which can be applied for via Capitec, FNB, and Standard Bank branches, as well as for EMV banking, transit ticketing, and building access.

Types of output devices
Varied computing devices delivering visual, audio, tactile, and physical outputs

Outputs are the results produced by a computer or electronic device, which can be visual, audio, printed, or specialized for accessibility.
Main Types of Outputs
1. Visual Output
Visual outputs display information in a form that can be seen. Common devices include monitors, projectors, and video cards. These devices show text, images, videos, or graphical data on screens, allowing users to interpret the information easily. Monitors can use technologies like LCD, LED, OLED, or CRT to present clear visuals

2. Audio Output

Audio outputs convert digital signals into sound. Devices such as speakers, headphones, and sound cards allow users to hear music, alerts, or spoken information. Audio output is essential for multimedia applications, virtual meetings, and accessibility features like text-to-speech

3.Printed Output

Printed outputs provide tangible results on paper or other physical media. Printers, plotters, and braille readers are examples. Printers can be inkjet, laser, or dot matrix, producing documents, images, or technical drawings. Braille displays help visually impaired users read text through raised dots

4.Specialized Output

Specialized outputs cater to specific needs or accessibility. Examples include Braille displays, tactile devices, and GPS devices. These outputs allow users to interact with information in non-standard formats, such as tactile or navigational feedback

1. Design output to serve the intended purpose - essentially, this means that the systems objectives should be met but the objectives should be linked to the business tactics

2. Design output to fit the user - design for the person, not the position; remember human information processing differences; issue of detail versus summary

3. Delivering the appropriate quantity of output - don't create overload, create enough to perform business tactics and make anything else optional

4. Making sure the output is where it is needed - right person at the right place; consider discontinuing printed output and see if anyone asks for it

5. Providing the output on time (and for the right time period)

6. Choosing the right output method- consider all senses (sound, smell, touch?)

REALIZING HOW OUPUT BIAS AFFECTS USERS

1. Bias when sorting information due to order affecting importance influencing recency effect.
2. Bias when limits are set, especially for exception reports.

3. Bias through graphics due to scales, size, color, or type of graphic. Viewing position also can introduce bias.

User/developer interaction and testing is the best way to avoid bias. Validity should be tested to assure that what you attempt to shown is what is perceived. Iteration is key since you won't get it right the first time.

SPECIFIC OUTCOME 3: Create computer input and output functions.
ASSESEMENT CRITERIA
❖ The creation ensures that the function format corresponds to the design. ❖ The creation ensures that the function behaviour corresponds to the design.

Activity	Question	SO	AC
3	**10. In groups describe how to create computer inputs and output functions. (10)**	3	1-2

Reading input from the user and showing it on the console (output) are the common tasks every C program needs. C language provides libraries (header files) that contain various functions for input and output. In this tutorial, we will learn different types of formatted and unformatted input and output functions.

The Standard Files in C
The C language treats all the devices as files. So, devices such as the "display" are addressed in the same way as "files".

The following three files are automatically opened when a program executes to provide access to the keyboard and screen.

To access a file, C uses a predefined FILE struct type to refer to the file for reading and writing purpose. Read this chapter to understand how to read values from the screen and how to print the result on the screen.

Types of Input and Output Functions
We have the following categories of IO function in C −

Unformatted character IO functions: getchar() and putchar()
Unformatted string IO functions: gets() and puts()
Formatted IO functions: scanf() and printf()

Output

```
Enter your name: Ravikant Soni
You entered the name: Ravikant Soni
```

scanf("%s") reads characters until it encounters a whitespace (space, tab, newline, etc.) or EOF. So, if you try to input a string with multiple words (separated by a whitespace), then it accepts characters before the first whitespace as the input to string.

fgets() Function

In newer versions of C, gets() has been deprecated as it is potentially a dangerous function because it doesn't perform bound checks and may result in buffer overflow.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

WEEK 6

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115367

Demonstrate logical problem solving and error detection techniques

Learner Information:

Details	Please Complete this Section
Name & Surname:	Simnikiwe Picane
Organisation:	PAICTA NSF
Unit/Dept:	
Facilitator Name:	Nomanesi Dinini
Date Started:	01 April 2026
Date of Completion:	08 July 2026

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OVERVIEW			
About the Learner Workbook	This Learner Exercise Workbook has been designed and developed to evaluate learners' level of understanding of the **Demonstrate logical problem solving and error detection techniques**		
Purpose	The purpose of this Learner Exercise Workbook is to evaluate learners understanding on the specific outcomes and/or assessment criteria of the following SAQA Registered Unit Standards:		
US No	**US Title**	**Level**	**Credits**
115367	**Demonstrate logical problem solving and error detection techniques**	**5**	**8**
Context	This assessment represents the Formative Assessment component of Function in a business environment, for the and should be completed in the classroom/training room.		
Resources	The following are resources needed for this assessment: 5. Learner Guide; and 6. Assessment Preparation.		
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the workbook to each learner; and • Interview the learner on similar questions, should he/ she not be able to write.		

Instructions to Learners	Learners will be required to: • Complete the workbook as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame of.... hours.
Total Mark	This formative assessment carries a total mark of _ **points**. In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.
Equipment	Learners are required to have the following equipment in order to complete this workbook: • Pen and Pencil; • Ruler; and • Exam Pad – for additional paper.

46.GENERAL INFORMATION

LEARNER DETAILS		
Learner Full Names	Simnikiwe Picane	
Learner ID No.:	0106181295085	
Organisation:	PAICTA NSF	
Unit/Dept:		
Contact Details:	**Telephone /Cell Numbers:**	**Email Address:**
	0710247715	Simnikiwepicane06@gmail.com
WORKSHOP DETAILS		
Workshop Venue:		
Facilitator Name:		
Date Started:		
Date Completed:		

47.ASSESSMENT PREPARATION CHECKLIST

DESCRIPTION	YE	N	COMMENTS/
This assessment is a formative assessment and it is based on the outlined unit standard/s for **Demonstrate logical problem solving and error detection techniques** module.			
Your assessment evidence **Demonstrate logical problem solving and error detection techniques** module needs to be submitted on..... (day) of.....(month).....(year) at the following address/			
You will be assessed based on the outlined Unit Standards. The assessment activities are linked to specific outcomes/assessment criteria of the outlined Unit Standards.			
To determine your competence level, the following are the methods to be used for this assessment: 5.			
To be declared competent on **Demonstrate logical problem solving and error detection techniques**module (formative assessment), you should have obtained at least 80% of the			
You will be provided with detailed feedback on your performance of this assessment as follows: 5. Written Feedback			
Should you be declared "not yet competent" on this assessment, you will be entitled for re-assessment opportunity/ies.			

You will be required to re-submit evidence (only for areas) you were declared not yet competent. A date for re-submission will be			
You will be entitled to lodge an appeal should you not be satisfied with the assessment decision of your assessment.			
You will be required to provide the assessor feedback on assessment procedure – this is to assist in improving the assessment practices.			
Your results of assessment and portfolio of evidence information will not be provided to any person without your written consent.			

Learner's Declaration			
I Simnikiwe Picane.....herewith declare that I am ready for the assessment, that we have reviewed the assessment preparation and plan, I understand the assessment process and I am happy that the assessment will be conducted in a fair manner.			
Learner Signature:	**Date:**	**Facilitator**	**Date:**

SESSION 1:

Describe different approaches to problem solving.

Learning outcome

- The description identifies the different problem solving techniques.
- The description identifies situations where specific problem solving techniques would be more suitable than others
- The description utilises the top-down problem solving approach in real life problems known to the learner.
- The description allows for the practice of problem breakdown in picture drawing applications.

Activity	P = Project Q = Questioning B = Brainstorming	Mark
1	**Distinguish between A bottom-up approach A top-down approach**	8

The top-down approach is characterized by a hierarchical structure where decisions are made at the highest level and communicated down through the organization. This approach is often used in traditional industries such as retail, healthcare, and manufacturing, where speed, efficiency, and consistency are prioritized. It creates a clear chain of command and authority, making it easier to manage and coordinate efforts across different departments. However, it may limit employee input and innovation, as decisions are made without much input or feedback from the lower levels of the organization.

In contrast, the bottom-up approach involves collaboration and participation from all levels of the organization, encouraging employees to share their ideas and feedback. This approach is more flexible and often used in industries that prioritize innovation, creativity, and empowerment. It fosters a more informal and decentralized culture, promoting teamwork and collaboration. However, it may slow down the decision-making process due to the need for consensus among all levels.
Simplilearn
+1

Both approaches have their own set of advantages and disadvantages, and the choice between them depends on the specific needs and goals of the organization. Balancing both approaches can help organizations harness the strengths of each and create a more effective and dynamic work environment.

SESSION 2: **Use logical operators in descriptions of rules and relationships in a problem situation.**
Learning outcome
• Usage describes the logical operators by drawing truth tables • Usage provides examples of problem situations where a specific operator can be used • Usage identifies which of the operators should be used to represent given situations.

Activity	P = Project Q = Questioning B = Brainstorming	Mark
2	**Draw these gates** **AND, OR, NOT, NAND, NOR, EXOR and EXNOR gates.**	10

electroniclinic.com

See More
Basic Logic Gates | Definition | Truth Tables | Examples
1. AND gate symbol.
2. This simple switching circuit simulates the operation of an AND gate. The lamp will not burn because both switches are off.
3. This is the same circuit as in Figure 2. However, the lamp will burn because both switches are on.
4. AND gate truth table. Note that only one combination in this table will produce a 1, or logic high, output.
5. Resulting valid logic highs and lows in an AND gate as they appear on a three-channel oscilloscope.

Electrical Academia
Do all digital logic gates follow their truth tables?
All logic gates follow their truth table. Digital logic gates are made to form RTL (resistor-transistor logic) or DTL (diode-transistor logic) circuits from discrete parts such as resistors, transistors, and diodes. The most “standard” personal digital logic gates along with their respective truth tables are described below.

theengineerspost.com
Logic Gates - AND,OR,NOT,NAND,NOR, XOR,XNOR | Truth Table | Digital logic design|Digital Electronics
15:55
Logic Gates - AND,OR,NOT,NAND,NOR, XOR,XNOR | Truth Table | Digital logic design|Digital Electronics
YouTubeSudhakar Atchala113.4K views21 Jan 2023

Including results for Draw these gates 2 AND, OR, NOT, NAND,

SESSION 3:

Simplify Boolean expressions with Boolean algebra and Karnaugh maps.

Learning outcomes

- The simplification describes the rules of Boolean algebra.
- The simplification uses the Boolean algebra rules to simplify given expressions.
- The simplification uses Karnaugh maps to represent Boolean expressions.
- The simplification involves writing down the simplified expression from the map.

Activity	P = Project	Mark
3	**Define** • **Boolean Expression/Function** • **K-Maps**	5

Boolean Expression/Function
A Boolean expression or function is a mathematical expression that represents binary logic operations. It consists of variables, logical operators (such as AND, OR, NOT), and constants (0 and 1). A Boolean function can be represented using truth tables or algebraic expressions.

K-Maps are graphical methods used to simplify Boolean expressions. They consist of a grid of cells, where each cell represents a unique combination of input variables. The cells are grouped together to identify patterns and simplify the Boolean expression.

SESSION 4:

Describe the basic concepts of error detection.

Learning outcome

- The description identifies the common causes of errors.
- The description identifies error isolation techniques.
- The description identifies various testing techniques.

Activity	P = Project	Mark
4	**Describe the basic concepts of error detection.**	5

Core Concepts of Error Detection

Error detection in computer networks ensures data integrity by identifying corruption during transmission. It works by adding redundant bits to the original data so the receiver can verify correctness. If errors are detected, the data is either discarded or retransmitted, depending on the protocol.

Types of Errors include:

Single-Bit Error – Only one bit is altered.

Multiple-Bit Error – Two or more bits are altered, not necessarily consecutively.

Burst Error – Several consecutive bits are corrupted, often due to noise or interference.

Key Error Detection Methods

Simple Parity Check Adds one bit to make the total number of 1s either even (even parity) or odd (odd parity). Detects all single-bit errors but fails for even-numbered bit errors. def even_parity(data): return data + str(data.count('1') % 2) print(even_parity("101010")) # Output: 1010101

S = SELF-ASSESSMENT CHECKLIST

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				
7				
8				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115373

Demonstrate an understanding of sort and search techniques used in computer programming

Unit Std #	115373
NQF Level	5
Notional hours	60
Credit(s)	6
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

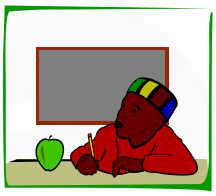
Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
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5				

Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				

4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



Activity	Practical	SO	AC
1	1. **In pairs list the advantages and disadvantages of using linked lists. (10)** 2. **Define Data type**	1	1-2

Advantages of Using Linked Lists
*Dynamic Size: Linked lists can grow or shrink at runtime, allowing for flexible memory allocation.

*Ease of Insertion/Deletion: Inserting or deleting nodes is straightforward, as only the pointers need to be updated.

*No Memory Wastage: Unlike arrays, linked lists do not waste memory space when elements are not used.

*Implementation of Data Structures: Linked lists can easily implement data structures like stacks and queues

Drawbacks of Linked Lists

Linked lists are flexible and efficient for insertion and deletion, but they come with several notable disadvantages that can impact performance and memory usage depending on the application.

1. Higher Memory Overhead Each node stores not only the data but also one or more pointers (to the next or previous node). This extra storage increases memory usage compared to arrays, especially for large datasets where pointer overhead becomes significant.

2. No Random Access Unlike arrays, linked lists do not support O(1) direct access by index. To reach the n-th element, you must traverse from the head node sequentially, resulting in O(n) access time.

3. Poor Cache Locality Array elements are stored contiguously in memory, enabling better CPU caching and prefetching. Linked list nodes are scattered in memory, leading to more cache misses and slower iteration.

4. Slower Traversal Traversal involves following pointers from node to node, which is slower than accessing contiguous memory in arrays. This also makes linked lists less suitable for GPU or SIMD-optimized operations.

5. Reverse Traversal Limitations Singly linked lists cannot be traversed backward without additional data structures. Doubly linked lists allow reverse traversal but require extra memory for back pointers.

6. Complex Implementation Managing pointers for insertion, deletion, and traversal increases implementation complexity and the risk of bugs such as memory leaks or dangling pointers.

7. Inefficient for Sorting Sorting linked lists is generally slower and more complex than sorting arrays. Algorithms like quicksort are less efficient due to non-contiguous storage; merge sort is preferred but involves pointer restructuring.

8. Not Ideal for Small Datasets For small collections, the overhead of pointers and traversal time outweighs the benefits of dynamic sizing, making arrays a better choice.

Example: Traversal Cost in Python

Activity	Practical	SO	AC
2	**12. In pairs distinguish between Selection sort and Insertion sort (5)**	2	1-2

Comparison of Selection Sort and Insertion Sort

Selection Sort and Insertion Sort are two fundamental sorting algorithms, each with distinct characteristics and use cases. While both divide the array into sorted and unsorted parts, their mechanisms for sorting differ significantly.

Selection Sort

Selection Sort repeatedly selects the smallest (or largest) element from the unsorted portion and swaps it with the leftmost unsorted element. This process ensures that the sorted portion grows incrementally from left to right.

*Time Complexity: It has a time complexity of $O(n^2)$ in all cases (best, worst, and average), as it always scans the unsorted portion to find the minimum element.

*Swaps: It performs fewer swaps compared to Insertion Sort, making it suitable when minimizing data movement is critical.

*Stability: It is an unstable algorithm, meaning it may not preserve the relative order of equal elements.

*Use Case: Selection Sort is useful for small datasets or when the cost of swapping is high, such as in flash memory.

Activity	Question	SO	AC
3	**11. In small groups distinguish between Basic Sequential Search and Binary Search. (5)**	3	1-2

In small groups distinguish between Basic Sequential 3 Search and Binary Search

In small groups, distinguishing between Basic Sequential Search and Binary Search involves understanding their fundamental differences and applications. Sequential Search is straightforward and can be implemented easily, but it is less efficient for large datasets due to its linear search approach. Binary Search, on the other hand, is more efficient for sorted datasets and can significantly reduce the search space with each comparison. Understanding these differences helps in choosing the appropriate search algorithm based on the data's size and structure.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

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Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
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Assessor	**Date**	**Moderator**	**Date**

WEEK 7

National Certificate: Information Technology (Systems Development)

SAQA ID: 44872

NQF Level5

Unit Standard: 114048

NQF L4

Credits: 2

Demonstrate an understanding of the handling of error in a computer

Unit Std #	115359
NQF Level	4
Notional hours	20
Credit(s)	2
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1 : Explain different errors found in the computer programming environment.
• The explanation covers the difference between an error and a mistake. • The explanation covers the errors found in data entry via computer input devices. • The explanation identifies sources of induced errors in calculations.
Specific Outcome 2 : Demonstrate how calculation errors are induced in the computer.
• The demonstration explains overflow errors found in computers. • The demonstration explains underflow errors found in computers. • The demonstration explains conversion errors found in computers. • The demonstration explains errors found in computers because of advancement in processor word-sizes.
Specific Outcome 3 : Demonstrate how mistakes and computer errors can be minimised
• The demonstration explains how mistakes can be minimised. • The demonstration explains how errors can be minimised.

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (2) x 10 = 20	6HRS	14HRS

1.	**Learning Programme Name:**			REFER TO COVER PAGE							
2.	**SAQA Qualification/ Unit Standard Title:**			REFER TO COVER PAGE							
3.	**Qualification/**		**4.**	**SAQA ID Number**		**5.**	**NQF Level**	4	**6.**	**Credits**	**2**
	Unit Standard										
7.	**<u>PURPOSE</u> for offering this programme to your learners:**			REFER TO NEXT PAGE							
8.	**<u>TARGET AUDIENCE</u> for this specific programme:**			REFER TO NEXT PAGE							
9.	**Entry/Admission Requirements:**			REFER TO NEXT PAGE							

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 6hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =14 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				

Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				
4				
5				

Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

2				
3				
4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: **Explain different errors found in the computer programming environment.**
ASSESEMENT CRITERIA
❖ The explanation covers the difference between an error and a mistake. ❖ The explanation covers the errors found in data entry via computer input devices. ❖ The explanation identifies sources of induced errors in calculations.

Activity	Practical	SO	AC
1	**1.1 In pairs Explain different errors found in the computer programming environment and differentiate them from mistakes. (10)**	1	1-3

Differentiating Errors from Mistakes

Errors are technical issues within the code that prevent it from running or cause it to behave unexpectedly.

Mistakes are conceptual or logical flaws in the design or implementation of the code, leading to incorrect behavior or results. In summary, errors are technical issues, while mistakes are more conceptual or logical in nature.

SPECIFIC OUTCOME 2 : Demonstrate how calculation errors are induced in the computer.
ASSESEMENT CRITERIA
❖ The demonstration explains overflow errors found in computers. ❖ The demonstration explains underflow errors found in computers. ❖ The demonstration explains conversion errors found in computers. ❖ The demonstration explains errors found in computers because of advancement in processor word-sizes.

Activity	Practical	SO	AC
2	**13. In pairs demonstrate understanding of how calculation errors are induced in the computer. (10)**	2	1-4

1. Floating-Point Precision Error

Concept: Computers store real numbers in binary floating-point format (IEEE 754), which cannot exactly represent all decimal fractions.

Example pair:

Calculation in exact arithmetic:

0.1

+

0.2

=

0.3

0.1+0.2=0.3

Computer calculation in floating-point:

0.1 + 0.2

Output:

0.30000000000000004

Explanation: 0.1 and 0.2 have repeating binary representations. Their sum introduces a tiny rounding error.

2. Round-Off Error in Iterative Calculations

Concept: Repeated arithmetic operations can amplify small floating-point errors.

Example pair:

Exact sum of 0.1 added ten times:

$0.1 + 0.1 + \dots + 0.1 = 1.0$

$0.1+0.1+\cdots+0.1=1.0$

2.Computer calculation:

```
sum([0.1]*10)
```

Output:

```
0.9999999999999999
```

Explanation: Each addition introduces a small round-off error, accumulating over iterations.

3.Truncation Error from Numeric Approximations

Concept: Computers approximate continuous functions (like derivatives) using discrete formulas.

Example pair:

True derivative:

$f'(x) = \cos(x), x = \pi/4$

$f'(x)=\cos(x),x=\pi/4$

SPECIFIC OUTCOME 3: Demonstrate how mistakes and computer errors can be minimised.
ASSESEMENT CRITERIA
❖ The demonstration explains how mistakes can be minimised. ❖ The demonstration explains how errors can be minimised.

Activity	Question	SO	AC
3	**12. In small groups discuss how mistakes and computer errors can be minimised. (5)**	3	1-2

Encourage Open Communication
In small groups, fostering an environment where members feel comfortable reporting mistakes or potential errors is essential. Open discussion allows the group to identify problems quickly and prevent them from escalating. Encourage team members to ask clarifying questions and verify instructions before proceeding.

Implement Standard Operating Procedures
Using checklists, step-by-step guidelines, and standard operating procedures (SOPs) reduces the likelihood of errors. Groups should ensure that everyone understands and follows the same procedures for data entry, software usage, and project tasks. Documented workflows provide a reference for consistency and accountability.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115362

Manage software development source files using appropriate tools

Unit Std #	115362
NQF Level	5
Notional hours	50
Credit(s)	5
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1: Locate software development source files.
• The location ensures that the correct source file is identified. • The location ensures that the correct versions of additional files associated with the software development source files are identified.
Specific Outcome 2 : Retrieve software development source files for update purposes.
• Retrieval and updating of the source file is in accordance with organisation procedures. • Retrieval protects source files from loss while updates are in progress. • Retrieval prevents source files from being updated simultaneously by two or more people.

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (5) x 10 = 50	15HRS	35HRS

1.	**Learning Programme Name:**			REFER TO COVER PAGE							
2.	**SAQA Qualification/ Unit Standard Title:**			REFER TO COVER PAGE							
3.	**Qualification/**		**4.**	**SAQA ID Number**		**5.**	**NQF Level**	5	**6.**	**Credits**	**5**
	Unit Standard										
7.	**PURPOSE for offering this programme to your learners:**			REFER TO NEXT PAGE							
8.	**TARGET AUDIENCE for this specific programme:**			REFER TO NEXT PAGE							
9.	**Entry/Admission Requirements:**			REFER TO NEXT PAGE							

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 15 hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =35 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				

Contact Details	**Email:**			
	Phone:		**Fax:**	

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- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
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5				

Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

2				
3				
4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: Locate software development source files.
ASSESEMENT CRITERIA
❖ The location ensures that the correct source file is identified. ❖ The location ensures that the correct versions of additional files associated with the software development source files are identified.

Activity	Practical	SO	AC
1	**1.1 In pairs discuss how to Locate software development source files (10)**	1	1-4

When discussing this workflow with your partner, consider breaking it down into four main phases:Repository Access: Determine if the project is open-source or private to figure out if you need public URLs, credentials, or VPN access.Navigation & Tracing: Use an Integrated Development Environment (IDE) like Visual Studio or IntelliJ IDEA to browse the directory structure, or apply Forward Tracing by setting breakpoints and logs to trace execution paths.Static & Dynamic Analysis: According to a consensus on Stack Exchange, developers often rely on static analysis tools like Doxygen to index code and map functions/classes Version History: Leverage development history and commit logs to identify which files were modified and who the domain experts are for those specific features.

SPECIFIC OUTCOME 2 :

Retrieve software development source files for update purposes.

ASSESEMENT CRITERIA

- ❖ Retrieval and updating of the source file is in accordance with organisation procedures.
- ❖ Retrieval protects source files from loss while updates are in progress.
- ❖ Retrieval prevents source files from being updated simultaneously by two or more people.

Activity	Practical	SO	AC
2	**14. In pairs discus the following, Testing Backup-and-Restore Procedures, Documenting Backup-and-Restore (10)**	2	1-3

Testing Backup and Restore Procedures

1. **Define Objectives**:
   - What data needs to be backed up?
   - What are the Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO)?

2. **Backup Types**:
   - Discuss different types of backups (full, incremental, differential).
   - Evaluate which method is most suitable for your scenario.

3. **Set Up a Testing Schedule**:
   - How often should backups be tested? (e.g., monthly, quarterly).
   - Consider using a test environment that mirrors the production environment for restoration tests.

4. **Perform Restore Tests**:
   - Document the process to restore data from backups.
   - Evaluate the time taken to restore data versus the RTO.
   - Verify the integrity of the data after restoration.

5. **Handling Exceptions**:
   - Discuss potential issues that might arise during the restore process.
   - How will you respond to failures or partial restores, and what troubleshooting procedures will you implement?

6. **Backup Media Verification**:
   - Discuss the importance of verifying the integrity of backup media.
   - How often will media be checked for failures or degradation?

Documenting Backup and Restore Procedures

1. **Procedure Documentation**:
   - Outline the step-by-step procedures for creating backups and restoring data.
   - Include screenshots or flowcharts for clarity.

2. **Roles and Responsibilities**:
   - Clearly define who is responsible for executing, monitoring, and testing backups.
   - Document who should be notified in case of a failure or issue.

3. **Access Controls**:
   - Document access permissions related to backup systems.
   - Ensure sensitive data is protected and access is limited to authorized personnel.

4. **Review and Update Procedures**:
   - Establish a schedule for reviewing and updating the backup and restore documentation.
   - Discuss how changes in technology or data landscape might require updates to procedures.

5. **Reporting and Logs**:
   - Discuss the importance of keeping logs of backup and restore operations.
   - How will these logs be reviewed and maintained?

6. **Training and Awareness**:
   - Document training sessions for relevant personnel on backup and restore procedures.
   - Ensure that staff are aware of the documented procedures and the processes for data restoration.

Conclusion
Wrap up the discussion by emphasizing the importance of regular testing and thorough documentation of backup and restore procedures. Highlight that these practices help mitigate risks associated with data loss and ensure quick recovery in case of incidents.

Action Points
Decide on specific actions to take next based on your discussion, such as drafting documentation, planning a testing schedule, or scheduling training sessions.

Feel free to explore these points with your partner, and remember that backup and restore processes are an ongoing commitment rather than one time tasks!

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

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LEARNING OUTCOMES		1	2	3
1				
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3				
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6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR

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Assessor	**Date**	**Moderator**	**Date**

WEEK 8

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 114049

Demonstrate an understanding of Computer Database Management Systems

Unit Std #	114049
NQF Level	5
Notional hours	70
Credit(s)	7
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1: Describe data management issues and how it is addressed by a DBMS.
• 1. The description identifies the problem they represent and includes examples. • 2. The description outlines ways which database management systems address the issues.
Specific Outcome 2 : Describe commonly implemented features of commercial database management systems
• 1. The description identifies the purpose of each feature. • 2. The description identifies the way in which each feature contributes to the solution of data management issues.
Specific Outcome 3 : Describe different type of DBMS`s.
• The description describes characteristics of the DBMS-type. • 2. The description gives examples of the use of the DBMS-type.
• **Specific Outcome 4 : Review DBMS end-user tools.**
• 1. The review identifies the features and limitations of the tools. • 2. The review outlines the interaction between the tools and the database. • 3. The review is based upon use of the tools.

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (7) x 10 = 70	21HRS	49HRS

1.	**Learning Programme Name:**			REFER TO COVER PAGE							
2.	**SAQA Qualification/ Unit Standard Title:**			REFER TO COVER PAGE							
3.	**Qualification/**		**4.**	**SAQA ID Number**		**5.**	**NQF Level**	5	**6.**	**Credits**	**7**
	Unit Standard										
7.	**<u>PURPOSE</u> for offering this programme to your learners:**			REFER TO NEXT PAGE							
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10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 21 hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =48 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
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	Phone:		**Fax:**	
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Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				

4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: Describe data management issues and how it is addressed by a DBMS.
ASSESEMENT CRITERIA
❖ 1. The description identifies the problem they represent and includes examples. ❖ 2. The description outlines ways which database management systems address the issues.

Activity		SO	AC
1	**1.1 What is a database management system (DBMS)? (2)** **1.2 In groups describe the main advantages of database management system (DBMS) (10)**	1	1-2

A database management class typically refers to an academic or training course focused on teaching the principles, tools, and practices of Database Management Systems (DBMS). Such a class covers how to design, implement, maintain, and optimize databases to store, retrieve, and manage data efficiently.

At its core, a DBMS is software that interacts with end users, applications, and the database itself to capture and analyze data. It ensures data integrity, security, and availability while supporting multiple users and applications simultaneously.

Advantages of Database Management System (DBMS)

A Database Management System (DBMS) provides numerous advantages that make it an essential tool for modern data management. One of the key advantages is data security.

DBMS enhances data security by implementing access control, encryption, and privacy policies. It prevents unauthorized access and ensures sensitive information remains secure. This is particularly important as the number of users accessing the system grows, reducing the risk of data breaches.

Another significant advantage is data consistency and accuracy. DBMS enforces integrity constraints, ensuring that updates are synchronized across all views of the data. This minimizes discrepancies, reduces errors, and guarantees reliable and consistent data.

Additionally, DBMS reduces data redundancy by eliminating duplicate entries through unique constraints. This not only optimizes storage usage but also improves data consistency by avoiding unnecessary repetition.

SPECIFIC OUTCOME 2 :

Describe commonly implemented features of commercial database management systems.

ASSESEMENT CRITERIA

- 1. The description identifies the purpose of each feature.
- 2. The description identifies the way in which each feature contributes to the solution of data management issues.

Activity		SO	AC
2	**15. In pairs describe commonly implemented features of commercial database management systems. (6)**	2	1-2

Commercial DBMSs typically implement features in complementary pairs, such as security with backup, and transaction management with concurrency control, to ensure data integrity, availability, and performance.

1. Data Security and Backup/Recovery
Data Security ensures that sensitive information is protected through user authentication, authorization, and encryption, preventing unauthorized access or tampering.

Backup and Recovery complements security by providing mechanisms for regular data backups and restoring the database in case of system failures or disasters, ensuring data availability and continuity

Together, these features protect both the confidentiality and durability of data.

2. Transaction Management and Concurrency Control
Transaction Management enforces ACID properties (Atomicity, Consistency, Isolation, Durability), ensuring that database operations are reliable and consistent
Concurrency Control manages simultaneous access by multiple users, preventing conflicts and maintaining data integrity through techniques like locking and transaction isolation.

These paired features allow multiple users to interact with the database safely without compromising correctness.

3. Query Optimization and Indexing

Query Optimization improves the efficiency of database queries by selecting the best execution plans, reducing response time.

Indexing supports this by creating data structures that speed up data retrieval operations. Together, they enhance performance and ensure fast access to large datasets.

4. Scalability and Data Replication

Scalability allows the database to handle increasing amounts of data and user load without significant performance degradation.

Data Replication complements scalability by synchronizing data across multiple locations, improving availability, fault tolerance, and load distribution. These features ensure that the system can grow with business needs while maintaining reliability.

5. Data Warehousing and Analytics Tools

Data Warehousing provides centralized storage for large volumes of historical data, supporting complex queries and reporting.

Analytics Tools or reporting features allow users to extract insights and make data-driven decisions. Paired together, they enable organizations to leverage stored data for strategic analysis and business intelligence.

These feature pairs illustrate how commercial DBMSs integrate complementary functionalities to provide secure, reliable, and high-performance database solutions for enterprise environments.

SPECIFIC OUTCOME 3: Describe different type of DBMS`s.
ASSESEMENT CRITERIA
❖ 1. The description describes characteristics of the DBMS-type. ❖ 2. The description gives examples of the use of the DBMS-type.

Activity	Question	SO	AC
3	**13. In small groups describe different type of DBMS`s (2)**	3	1-2

In-Pairs Groups Describe Different Types of DBMS's

In pairs groups can discuss the different types of DBMSs by providing examples and discussing their use cases. For instance, they can explain how a Hierarchical DBMS organizes data in a tree-like structure, suitable for organizational charts, and how a Network DBMS allows for complex relationships, often used in telecommunications. They can also discuss the advantages and disadvantages of each type, such as the flexibility of a Relational DBMS and the scalability of a NoSQL DBMS.

SPECIFIC OUTCOME 4: Review DBMS end-user tools.
ASSESEMENT CRITERIA
❖ 1. The review identifies the features and limitations of the tools. ❖ 2. The review outlines the interaction between the tools and the database. ❖ 3. The review is based upon use of the tools.

Activity	Question	SO	AC
4	**7. In pairs describe the Relationship between Visual Database Tools and Databases (8)**	4	1-3

Visual Database Tools and Databases

Visual Database Tools are graphical interfaces that allow developers and database administrators to design, manage, and interact with databases without relying solely on manual SQL scripting. They provide a bridge between the conceptual design of a database and its physical implementation.

1. Database Structure Representation ↔ Visual Modeling Databases store structured data in objects like tables, columns, keys, indexes, relationships, constraints, and views. Visual Database Tools, such as Database Designer, represent these objects in graphical diagrams, making it easier to understand and modify the database schema. For example, the Database Designer in SQL Server Management Studio lets you visually create and edit relationships between tables, which directly updates the underlying database schema.

2. Object Creation and Modification ↔ Interactive Design Tools While databases require SQL commands to create or alter objects, Visual Database Tools offer drag-and-drop interfaces and property panels to perform these actions. Tools like Table Designer allow detailed table creation with column definitions, data types, and constraints, while View Designer simplifies creating database views by visually building queries instead of writing raw SQL.

-Example: Creating a Table Visually
SQL equivalent of a visual table creation
CREATE TABLE Customers (
CustomerID INT PRIMARY KEY,
Name NVARCHAR(100),
Email NVARCHAR(100) UNIQUE

Copy
In a visual tool, this would be done by adding columns in a grid, setting data types from dropdowns, and marking constraints via checkboxes.

3. Query and View Design ↔ SQL Generation Databases execute queries in SQL, but Visual Database Tools like the Query and View Designer let users build queries by selecting tables, choosing columns, and applying filters visually. The tool then generates the SQL code automatically, ensuring accuracy and saving time.

4. Schema Maintenance ↔ Diagram-Based Updates When a database schema changes, visual tools allow real-time updates through diagrams. Modifications made visually are translated into SQL ALTER statements, keeping the database in sync with the design.

In short, Visual Database Tools act as an intuitive layer over databases, enabling faster design, easier maintenance, and reduced syntax errors while still producing valid SQL commands behind the scenes.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

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3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
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6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

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Assessor	**Date**	**Moderator**	**Date**

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 114048

Create database access for a computer application using structured query language

Unit Std #	114048
NQF Level	5
Notional hours	90
Credit(s)	9
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Contact Details

Standard Title:	

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Control sheet

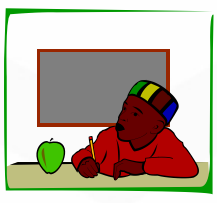
Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



Activity	Practical	SO	AC
1	**1.1 In pairs describe Abstraction and Data Independence Types. (10)**	1	1-4

*Physical Level: Describes how a record is actually stored in the database.

*Logical Level: Describes the actual data stored in the database.

*View Level: Allows users to view data in rows and columns.

*Physical Data Independence: The ability to change the physical storage of data without affecting the logical schema.

*Logical Data Independence: The ability to modify the logical schema without affecting the external views or applications.

Activity	Practical	SO	AC
2	**16. In pairs demonstrate understanding of Simple indexes Queries and Index Structures. (5)**	2	1-4

Simple indexes are data structures that allow efficient retrieval of records from a database table by organizing key column values for quick lookups.

What is a Simple Index?

A simple index is a database structure that stores values from a specific column (or columns) along with pointers to the corresponding rows. This allows the database to quickly locate records without scanning the entire table, improving query performance

The column(s) used in the index are called search keys.

Common Index Structures

B-Tree Index

A balanced tree structure with a root, intermediate nodes, and leaf nodes.

Efficient for range queries and equality searches.

Example: Retrieving all records where age = 30 can use a B-Tree index on the age column to quickly locate matching rows

Hash Index

Uses a hash function to map values to specific locations.

Very fast for equality searches but not suitable for range queries.

Example: Finding all records where city = 'New York' can use a Hash index on the city column

Bitmap Index

Uses a bit for each distinct value in a column.

Efficient for filtering columns with low cardinality (few distinct values).

Example: Retrieving all records where gender = 'Male' can use a Bitmap index to quickly identify matching rows

Types of Indexes in SQL

Primary Index: Automatically created on the primary key; ensures uniqueness and fast access.

Secondary Index: Created on non-primary key columns that are frequently queried.

Composite Index: Covers multiple columns to optimize queries filtering on more than one column

Example Queries Using Indexes

Single Column Index (B-Tree)

CREATE INDEX idx_age ON Customers(age);

SELECT * FROM Customers WHERE age = 30;

The B-Tree index allows the database to traverse the tree and retrieve rows efficiently

Hash Index for Equality Search

CREATE INDEX idx_city ON Customers(city);

SELECT * FROM Customers WHERE city = 'New York';

The hash function maps 'New York' to a specific location, speeding up retrieval

Activity	Question	SO	AC
3	14. In small groups discuss User-Defined Routines and User-defined types (UDTs). (5)	3	1-8

Certainly! Let's break down the concepts of User Defined Routes (UDR) and User Defined Types (UDT) in the context of computer science and software development.

User Defined Routes (UDR)

Definition:
User Defined Routes (UDR) are custom routing configurations that allow developers to specify how requests to a web application should be handled based on certain parameters, such as URLs or HTTP methods.

Key Points:
1. **Custom Logic:** UDRs enable you to define custom logic that determines how different types of requests are routed to appropriate controllers or functions.
2. **Flexibility:** They provide more flexibility than the built-in routing mechanisms and allow for advanced patterns (e.g., nested routes, dynamic segments).
3. **Use Cases:** Commonly used in web frameworks such as Express (Node.js), Flask (Python), and Ruby on Rails to create RESTful APIs or web applications.
4. **Parameters:** UDRs can handle parameters, allowing dynamic data to be passed through the URL (e.g., `/users/:id`).
5. **Middleware Integration:** Often used in integration with middleware to perform actions such as authentication or logging before the request reaches the intended route handler.

User Defined Types (UDT)

Definition:
User Defined Types (UDTs) allow developers to create their own data types in programming languages, which enables better organization and management of data structures.

Key Points:
1. **Encapsulation:** UDTs allow encapsulating related data and operations into single entities, improving code readability and maintainability.
2. **Types of UDTs:**
   - **Structures (structs):** A collection of related fields that can vary in type (used in languages like C/C++).
   - **Classes:** Object-oriented programming structures that define properties (attributes) and methods (functions) for creating objects (found in languages like Python, Java, and C#).
   - **Enumerations (enums):** A distinct type that consists of a set of named values, enhancing code clarity (used in languages like C and Java).
3. **Database Context:** In databases, UDTs allow the creation of complex types for columns in tables. They support structured data types like address or phone numbers.

Activity	Question	SO	AC
4	1. Describe typical debugging process. (5)	4	1-2

Typical Debugging Process

The typical debugging process involves several key steps that help identify, isolate, and fix issues in software or code. Here's a breakdown of the steps:

Identify the Problem: Reproduce the bug, gather information, and understand the system.

Analyze the Code: Review the relevant code segments and familiarize yourself with the logic and flow.

Formulate Hypotheses: Identify potential causes based on your understanding.

Test Hypotheses: Write tests targeting your hypotheses and use debugging tools to step through the code.

Fix the Issue: Modify the code to resolve the bug and ensure correct functionality.

Test and Validate: Thoroughly test the application to confirm that the bug is resolved and that the fix has not adversely affected other parts of the system.

This systematic approach ensures that bugs are found, analyzed, and fixed effectively, leading to a more stable and reliable software product.

Activity	Question	SO	AC
5	1. **Demonstrate understanding of Documenting programs for a computer application that accesses a database using SQL. (5)**	5	1-2

Documenting Programs for Applications Accessing Databases Using S…

Documenting programs that interact with databases is essential for ensuring maintainability, clarity, and collaboration among developers and stakeholders. Proper documentation provides a clear understanding of the database structure, relationships, and the logic implemented in the application.

Key Aspects of Documentation

1. Code-Level Documentation

Embed comments in SQL scripts and application code to describe the purpose of tables, columns, constraints, and queries. For example:

```
-- Table: users
CREATE TABLE users (
user_id INT PRIMARY KEY AUTO_INCREMENT, -- Unique identifier for each user
username VARCHAR(50) UNIQUE NOT NULL, -- User's unique username
email VARCHAR(100) UNIQUE NOT NULL, -- User's email address
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP -- Record creation timestamp
);
```

Copy

This ensures that developers can quickly understand the purpose of each database object.

2. Entity-Relationship Diagrams (ERDs)

Use tools like MySQL Workbench or Lucidchart to create visual representations of the database schema. ERDs should include tables, columns, relationships, and constraints, making it easier to understand the database's structure.

3. Stored Procedures and Functions Documentation

Document stored procedures and functions by including their purpose, input/output parameters, and logic. For example:

```
DELIMITER //
CREATE PROCEDURE getUserDetails (
IN userId INT,
OUT username VARCHAR(50),
OUT email VARCHAR(100)
)
BEGIN
SELECT username, email INTO username, email
FROM users
WHERE user_id = userId;
END //
DELIMITER ;
```

This helps developers understand the encapsulated business logic and its impact on database behavior.

4. Extended Properties for SQL Server

For SQL Server, use Extended Properties to store metadata about database objects. For instance:

```
EXEC sp_addextendedproperty
@name = N'MS_Description',
@value = N'Table storing user information',
@level0type = N'SCHEMA', @level0name = 'dbo',
@level1type = N'TABLE', @level1name = 'users';
```

Copy

This metadata can be accessed programmatically or through SQL Server Management Studio (SSMS).

5. Version Control

Store SQL scripts and documentation in version control systems like Git. This ensures changes to the database schema are tracked and can be reverted if necessary

6. Regular Updates

Update documentation whenever changes are made to the database schema, such as adding new tables, modifying columns, or updating constraints. This ensures the documentation reflects the current state of the database.

7. Missing Documentation Identification

Use queries or tools to identify undocumented database objects. For example, create a view to list objects missing descriptions:

```
SELECT
TABLE_NAME,
COLUMN_NAME
FROM INFORMATION_SCHEMA.COLUMNS
WHERE COLUMN_COMMENT IS NULL;
```

Copy

This helps ensure all objects are properly documented.

8. Automated Documentation Tools

Leverage tools like ApexSQL Doc or custom scripts to generate documentation automatically. These tools can extract metadata and create comprehensive reports in formats like HTML or PDF.

Benefits of Proper Documentation

Facilitates onboarding of new developers by providing clear insights into the database and application logic.

Reduces maintenance overhead by making it easier to understand and modify existing code.

Enhances collaboration among team members by providing a shared understanding of the database structure and application logic.

By following these practices, you can ensure that your programs and databases are well-documented, maintainable, and aligned with best practices.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

S.Picane 08 July 2026

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR

I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

WEEK 9

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115392

Apply principles of creating computer software by developing a complete programme to meet given business specifications

Unit Std #	115392
NQF Level	5
Notional hours	12
Credit(s)	120
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

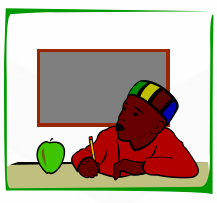
Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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2				
3				

4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



Activity	Practical	SO	AC
1	1. **In pairs Explain Why Should Technical Writers be Involved with Software Requirements Specifications?. (10)** 2. **What Kind of Information Should an SRS Include?**	1	1-3

Technical writers should be involved with Software Requirements Specifications (SRS) because they play a vital role in ensuring the SRS is clear, user-focused, consistent, and serves as a bridge between technical and non-technical stakeholders. They help identify gaps, maintain documentation standards, and advocate for user needs, which are crucial for project success. Their detailed approach ensures that all stakeholders have a shared understanding of the project scope and that the SRS is accessible and understandable.

Activity	Practical	SO	AC
2	**17. In pairs demonstrate understanding of how to establish Business Rules for Contingencies and Responsibilities and Establish a Requirements Traceability Matrix (10)** **18. In small groups discuss about Design considerations**	2	1-4

Building an RTMTo create a basic Requirements Traceability Matrix on Creately, set up a table/spreadsheet with the following core components:Requirement ID: A unique, unchangeable identifier for each project requirement (e.g., REQ-01).Requirement Description: A brief, clear summary of what the stakeholder or system needs.Design ID / Deliverable Link: A cross-reference to the Work Breakdown Structure (WBS) or visual design specification addressing the requirement.Test Case Reference: The specific unit, integration, or User Acceptance Testing (UAT) case verifying the requirement.Status & Owner: The current state (e.g., pending, in design, tested) and the designated team member accountable.

When discussing Design Considerations in your small group, focus on how requirements and contingencies directly shape the physical or architectural structure of the project:Feasibility & Technical Constraints: Are the requirements technically achievable within the established budget and technology stack?Scalability & Flexibility: Does the design allow for contingencies, and can it accommodate future modifications without breaking the traceability chain?Risk Management & Failure Modes: How does the design handle system failures, and are the contingency business rules built directly into the workflow logic?Testing & Compliance Validation: Is every design element easily testable to fulfill mandatory regulatory or quality standards, as mapped out in the RTM?Would you like to see a step-by-step example scenario of an 'if-then' contingency business rule, or a blank RTM table template you can copy for your project?10 sitesWhat Is a Requirements Traceability Matrix? (With Examples ...28 Aug 2025 — Standardize columns:Include Requirement ID, description, source, related test cases, verification method, status, and owner. Consi...Stell EngineeringHow to Make a Requirements Traceability Matrix (RTM)27 Jan 2025 — Who Makes the Requirements Traceability Matrix? Multiple stakeholders typically collaborate to gather the requirements for an RTM.ProjectManagerThe Correct Requirements Traceability Matrix [+template]05 May 2020 — but only partially and from their point of view the RTM that we project managers use is a different tool that combines. the best .

Activity	Question	SO	AC
3	**15. In small groups discuss how Create a computer program that implements the design. (5)**	3	1-2

1.Understand the Design
*Begin by reviewing the design thoroughly. This may include flowcharts, UML diagrams, pseudo-code, or written requirements.
*Make sure everyone in the group understands the program's purpose, expected inputs, outputs, and key functionalities.
2. Break Down the Problem
Divide the program into modules or components: Each member can take responsibility for a specific part, such as a user interface, data processing, or database connection.
Identify dependencies: Determine which modules must be implemented first and how they will interact.

Discuss algorithms: Decide on the logic or methods needed to solve each component. For example, sorting, searching, or mathematical calculations.

3. Choose the Programming Environment and Language
Select a programming language suitable for the task (e.g., Python for general-purpose, Java for object-oriented programs).
Agree on tools such as code editors, IDEs, or version control systems like Git.

4. Implementation Plan
Assign tasks: Allocate specific tasks to group members based on skills and interests.
Write pseudo-code or skeleton code: Helps to visualize logic before full implementation.
Establish coding standards: Consistent naming conventions, indentation, and comments ensure readability and maintainability.

5.Collaborative Coding Practices
Use version control (e.g., Git) so members can work on different parts and merge changes safely.
Conduct pair programming or peer reviews to reduce errors and improve code quality.
Keep regular communication, using meetings or chat tools, to track progress and resolve issues early.

6.Testing and Integration
Test individual modules first (unit testing) before combining them.
Conduct integration testing to check how all parts work together.
Fix bugs collaboratively and document solutions for future reference.
7. Documentation and Presentation
Document code with comments, usage instructions, and a brief description of each module.
Prepare to present your program to the class or instructor, explaining design decisions, challenges, and solutions.

Example Approach in Python (Group Task)

```
# Example: small task division
# Module 1: input numbers
def get_numbers():
    return [int(x) for x in input("Enter numbers separated by space: ").split()]
```

Activity	Question	SO	AC
4	**1. In small groups demonstrate understanding of how to Test a computer program. (10)**	4	1-2

To test a computer program effectively, you can follow these steps:

Planning
Understand Requirements: Review the program's specifications to understand its expected behavior.
Identify Test Cases: Create test cases that cover different scenarios, including normal inputs, boundary conditions, and erroneous inputs.
Writing Tests
Unit Testing: Write tests for individual components or functions to ensure they work as expected.
Integration Testing: Test the interaction between different components to verify their combined functionality.
System Testing: Test the entire system to ensure it meets the requirements.
Execution
Automated Testing: Use testing frameworks to automate the execution of test cases, saving time and ensuring consistency.
Manual Testing: Perform manual tests to simulate real user interactions and identify any unexpected behavior.

Analysis
Identify Defects: Document any failures or unexpected behavior observed during testing.
Regression Testing: Re-run tests to ensure that fixes for identified defects do not introduce new issues.
Documentation
Test Reports: Create reports documenting the test results, including any defects found and their resolutions.
By following these steps, you can systematically test a computer program to ensure its reliability and functionality.

Activity	Question	SO	AC
5	1. **In pairs describe how to implement the program to meet business requirements. (5)**	5	1-2

Implementing a program to meet business requirements involves several key steps. Here's a structured approach:

1. **Requirements Gathering**
- **Identify Stakeholders:** Engage with all relevant stakeholders, including business leaders, end-users, and technical teams.
- **Conduct Interviews and Workshops:** Use interviews, focus groups, and workshops to elicit detailed requirements.
- **Document Business Requirements:** Record all the gathered requirements clearly, ensuring they are specific, measurable, achievable, relevant, and time-bound (SMART).

2. **Analysis and Planning**
- **Analyze Requirements:** Prioritize the requirements based on business needs and impacts.

- **Create Use Cases and User Stories:** Develop use cases or user stories to outline how users will interact with the system.
- **Draw Flowcharts and Diagrams:** Visual representations of processes, data flow, and user interactions can be helpful.

3. **Design**
- **System Architecture:** Define the architecture of the program, including data stores, user interfaces, and integration points with other systems.
- **Technical Specifications:** Create detailed technical specifications that guide developers in building the system.
- **Prototype Development:** If applicable, develop prototypes to validate designs with stakeholders before full-scale development.

4. **Development**
- **Set Up Development Environment:** Configure the tools and environments needed for development.
- **Agile/Scrum Methodology:** Consider iterative approaches like Agile or Scrum to develop the program in sprints, allowing for regular feedback.
- **Coding Standards:** Establish coding standards and best practices to ensure quality and maintainability.

5. **Testing**
- **Test Plan Creation:** Develop a test plan that outlines the testing approach, including unit testing, integration testing, system testing, and user acceptance testing (UAT).

- **Automated and Manual Testing:** Employ both automated and manual testing strategies to ensure comprehensive coverage.
- **User Acceptance Testing:** Engage end-users to validate the application against business requirements.

6. **Deployment**
- **Deployment Strategy:** Define how the application will be deployed (phased, big bang, etc.) and create a rollback plan in case of issues.
- **Training and Documentation:** Prepare user manuals and training sessions for end-users and administrators.
- **Data Migration:** Plan and execute data migration if necessary.

Activity	Question	SO	AC
6	1. **Write short notes on how to Document the program according to industry standards.. (5)**	6	1-2

Documenting a Program According to Industry Standards
Comments: Use comments to explain complex code, especially for non-obvious logic or algorithms. Comments should be clear, concise, and written in plain language.
Function and Variable Names: Use descriptive names for functions and variables to make the code self-explanatory. Avoid abbreviations and acronyms unless they are widely understood in the industry.

Documentation Tools: Utilize industry-standard documentation tools such as Doxygen, Javadoc, or Sphinx to generate professional-looking documentation from code comments.
Code Structure: Follow a consistent code structure and formatting style. Use indentation, spacing, and consistent naming conventions to enhance readability.
Version Control: Use version control systems like Git and include meaningful commit messages to track changes and provide context for modifications.
User Manuals: If the program is intended for end-users, provide a user manual that explains how to install, configure, and use the software.
API Documentation: If the program includes an API, thoroughly document the endpoints, request/response formats, and usage examples for developers.
Change Logs: Maintain a change log to document updates, bug fixes, and new features. This helps users and other developers understand the evolution of the program.
License and Legal Documentation: Include the appropriate license information and any legal disclaimers or notices required for the software.
By adhering to these industry standards, you can ensure that your program is well-documented, maintainable, and accessible to other developers and users.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

WEEK 10

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115384

Test a computer program against a given specification

Unit Std #	115384
NQF Level	5
Notional hours	60
Credit(s)	6
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1 : Test a computer program against given specifications according to test plans.
• The testing executes operational steps identified in the test plan. • The testing uses input data as specified in the test plan. • The testing outlines deviations from the test plan, with explanations. • The testing follows the standards and procedures specified in the test plan for testing and re-testing.
Specific Outcome 2 : Record the results from testing a computer program
• The records are provided for all tests executed. • The records identify variations from expected test results and gives reason where available. • The recorded results are reproduced if the tests are repeated under the same conditions. • The recorded results are recorded in a way that allows the results to be reviewed..
Specific Outcome 3 : Review the testing process for a computer program against organisation policy and procedures
• The review allows improvements to be made to the application testing process. • The review follows organisation policy and procedures.

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (6) x 10 = 60	15HRS	35HRS

1.	**Learning Programme Name:**	REFER TO COVER PAGE									
2.	**SAQA Qualification/ Unit Standard Title:**	REFER TO COVER PAGE									
3.	**Qualification/ Unit Standard**		**4.**	**SAQA ID Number**		**5.**	**N Q F Level**	5	**6.**	**Credits**	**6**
7.	**PURPOSE for offering this programme to your learners:**	REFER TO NEXT PAGE									
8.	**TARGET AUDIENCE for this specific programme:**	REFER TO NEXT PAGE									
9.	**Entry/Admission Requirements:**	REFER TO NEXT PAGE									

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 15 hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =35 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				
4				
5				

Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				
2				
3				

4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: **Test a computer program against given specifications according to test plans.**
ASSESEMENT CRITERIA
❖ The testing executes operational steps identified in the test plan. ❖ The testing uses input data as specified in the test plan. ❖ The testing outlines deviations from the test plan, with explanations. ❖ The testing follows the standards and procedures specified in the test plan for testing and re-testing.

Activity	Practical	SO	AC
1	**1.1 In groups discusses how to Test a computer program against given specifications according to test plans. (10)**	1	1-4

Step 1 — Understand the Test Plan and Specifications

Start by carefully reviewing the software specifications and the test plan. The test plan typically outlines:

Objectives of testing

Scope and boundaries of the tests

Features to be tested and not tested

Test environment setup

Required resources, personnel, and timelines

Testing methods (manual or automated)

Ensure you understand both functional requirements (what the software should do) and non-functional requirements (performance, usability, security, etc.).

Step 2 — Design Test Cases

Based on the specifications, create test cases for each functional requirement. Each test case should include:

Test identifier and description

Preconditions (state of the system before the test)

Input data and conditions

Expected outcome based on the specifications

Steps to execute the test

It is also helpful to design positive tests (checking correct behavior) and negative tests (checking error handling or invalid input).

Step 3 — Set Up the Test Environment

Prepare the testing environment to match production or specified conditions. This includes installing the software version to be tested, configuring hardware and software settings, and ensuring that testing tools or scripts are ready.

Step 4 — Execute Tests
Run the test cases according to the test plan. For each test:
Input the specified data and perform the outlined steps
Observe and record the actual results
If using automated testing, execute the test scripts and capture results automatically; for manual testing, carefully log both successes and discrepancies.
Step 5 — Compare Actual vs Expected Results
For each test case, compare the actual behavior of the program with the expected outcome defined in the specifications.
Pass if the actual result matches the expected result
Fail if there is a deviation
Document any failed test cases for further analysis and bug tracking.

Step 6 — Report and Track Defects
For any discrepancies or failures:
Record defects clearly, including steps to reproduce, input data, and observed results
Assign severity and priority according to the impact on functionality
Track defect resolution and retest once issues are fixed
Step 7 — Regression and Re-Testing
After defects are corrected, re-test affected areas to ensure compliance with specifications and verify that new changes have not introduced additional issues (regression testing).

Step 8 — Test Summary and Sign-Off
Compile a test summary report including:
Number of test cases executed, passed, and failed
List of defects and their resolutions
Observations on compliance with specifications
This report provides evidence that the program has been verified against the given specifications and supports final approval or release.
By following these steps, testing can effectively ensure that the program behaves as intended and meets all documented requirements.

SPECIFIC OUTCOME 2 :

Record the results from testing a computer program.

ASSESEMENT CRITERIA

- ❖ The records are provided for all tests executed.
- ❖ The records identify variations from expected test results and gives reason where available.
- ❖ The recorded results are reproduced if the tests are repeated under the same conditions.
- ❖ The recorded results are recorded in a way that allows the results to be reviewed..

Activity	Practical	SO	AC
2	**19. In pairs outline how to Record the results from testing a computer program (10)**	2	1-4

Pair-Testing Documentation WorkflowSetup & Roles:Driver: Operates the program, inputs test data, and reads back system responses.Navigator: Verifies that the actual outputs match the requirements and types documented in your Test Summary Report guidelines.Executing and Logging (Step-by-Step):Note the exact Date, Time, and Software/Build Version being tested.For every individual test, fill out a shared Testing Status Report table containing:Test Case ID: A unique identifier (e.g., TC-01).Test Description: What specific feature or module you are validating.Input / Action: The exact keystrokes, clicks, or data entered into the program.Expected Result: What the program is supposed to do.Actual Result: What the program actually did (note any error messages or crashes).Status: Mark clearly as Pass, Fail, or Skipped.Handling Failures and Bugs:If a test fails, the pair should pause and agree on the defect.Draft a concise Defect Report outlining the anomaly, the steps to reproduce it, and attach any relevant error logs or screenshots.Would you like me to provide a downloadable spreadsheet template format for your pair-testing logs, or do you need strategies on how to test edge cases?10 sitesHow do you write good test reports for software testing?04 Apr 2025 — A test report summarizes the results of testing a software application. It includes details about: * **Test objectives** * **Test ...DevinitiTest Report in Software Testing: Guide, Format & Examples - testomat.io16 Mar 2026 — A test report typically includes the following components: * **Test execution summary and recommendations** The report should be c...Testomat.ioSoftware System Test Report - FormlyAI DocumentationQ&A. Document each test failure with complete details including the specific test case, expected vs. actual results, investigation...docs.formly.aiShow all

3.Handling Failures and Bugs:If a test fails, the pair should pause and agree on the defect.Draft a concise Defect Report outlining the anomaly, the steps to reproduce it, and attach any relevant error logs or screenshots.

SPECIFIC OUTCOME 3:

Review the testing process for a computer program against organisation policy and procedures.

ASSESEMENT CRITERIA

- ❖ The review allows improvements to be made to the application testing process.
- ❖ The review follows organisation policy and procedures.

Activity	Question	SO	AC
3	**16. In groups describe Different types of reviews (10)**	3	1-2

Research reviews synthesize existing scholarly literature to establish foundations, map evidence, and identify gaps. The primary types are:
*Narrative Review: Summarizes and critiques broad topics based on the author's selection, making it flexible but prone to bias.
*Systematic Review: Uses strict, reproducible protocols and predefined criteria to minimize bias and answer highly focused questions.
Scoping Review: Maps the breadth of available evidence or emerging topics without formal quality appraisal.Meta-Analysis: A statistical technique often paired with systematic reviews that combines quantitative data from multiple studies to calculate an overall effect size.

*Umbrella Review: An overview that synthesizes multiple existing systematic reviews on related questions in the same topic area.Integrative Review: Combines diverse methodologies (both qualitative and quantitative data) to create a holistic understanding of a complex topic.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I ……..... (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115388

Produce documentation for a computer program to agreed standards

Unit Std #	115388
NQF Level	5
Notional hours	30
Credit(s)	3
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1: Plan and design documentation for a computer program to agreed standards.
• The documentation design covers the format of the documents in line with industry conventions. • The documentation plan covers full program documentation components.
Specific Outcome 2: Create documentation for a computer program to agreed standards.
• The documentation is created according to the design created in specific outcome 'Plan and design documentation for a computer program to agreed standards'. • The documentation components created are created according to the plan specified in specific outcome 'Plan and design documentation for a computer program to agreed standards'. • The documentation created is structured sensibly, defining how program specifications have been met.
Specific Outcome 3: Review documentation for a computer program for completeness.
• The review identifies if documentation was created according to the design created in specific outcome 'Plan and design documentation for a computer program to agreed standards'. • The review identifies if documentation was created consistent with the computer program being documented.

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (5) x 10 = 50	9HRS	21HRS

1.	**Learning Programme Name:**			REFER TO COVER PAGE							
2.	**SAQA Qualification/ Unit Standard Title:**			REFER TO COVER PAGE							
3.	**Qualification/**		**4.**	**SAQA ID Number**		**5.**	**NQF Level**	5	**6.**	**Credits**	**3**
	Unit Standard										
7.	**<u>PURPOSE</u> for offering this programme to your learners:**			REFER TO NEXT PAGE							
8.	**<u>TARGET AUDIENCE</u> for this specific programme:**			REFER TO NEXT PAGE							
9.	**Entry/Admission Requirements:**			REFER TO NEXT PAGE							

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 9 hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =21 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

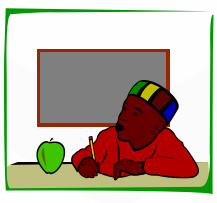
Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: Plan and design documentation for a computer program to agreed standards.
ASSESEMENT CRITERIA
❖ The documentation design covers the format of the documents in line with industry conventions. ❖ The documentation plan covers full program documentation components.

Activity	Practical	SO	AC
1	**1.1Describe the Role of documentation in software development (6)** **1.2 In groups distinguish between Process and Product Documentation (10)** **1.3 List Examples of product standards which should be developed (10)**	1	1-2

The main roles of documentation break down into distinct areas across the development lifecycle:Team Alignment & Communication: Acts as a blueprint that conveys requirements and design choices between stakeholders, developers, and product managers Studocu, YouTube.Knowledge Transfer & Onboarding: Reduces the learning curve for new hires, letting them get up to speed independently without draining senior developers' time Prestious, Crest Infotech.Maintenance & Debugging: Code comments, API references, and architecture diagrams provide a roadmap for debugging old logic and extending features safely Brightmarbles, TechTarget.End-User Enablement: Manuals, tutorials, and how-to guides empower users and IT staff to operate the system, which deflects support tickets TechTarget, Medium.Compliance & Governance: Provides mandatory audit trails to satisfy regulatory requirements in highly sensitive industries like healthcare and finance Crest Infotech, Zaycodes.

Safety Standards: Protect consumers from harm, such as electrical guidelines for household appliances Studocu or Machinery Directives.Interoperability Standards: Ensure seamless communication between hardware and software, such as Bluetooth connectivity or USB protocols Studocu.Environmental Standards: Address sustainability metrics like energy efficiency limits in electronic devices Studocu or RoHS (Restriction of Hazardous Substances).Quality Management Standards: Frameworks like ISO 9001 that define consistent production quality.Labeling Standards: Regulate transparent product disclosures including nutritional values or chemical warnings Studocu.Performance Standards: Define minimum thresholds for speed, durability, and reliability Studocu, Productboard.

Product standards ensure quality, safety, and interoperability across industries. Core categories that should be developed include Safety Standards (e.g., Electrical Safety Standards for appliances), Environmental Standards (e.g., energy efficiency limits), Interoperability Standards (e.g., universal connectivity like Bluetooth), and Performance Standards (e.g., ASTM Materials benchmarks).If you would like, let me know:A specific industry or product type (e.g., electronics, medical devices, software)Whether you are looking for compulsory regulations versus voluntary guidelinesI can narrow down the exact compliance requirements and Product Requirements Document (PRD) standards needed for your specific use case.

SPECIFIC OUTCOME 2 :

Create documentation for a computer program to agreed standards.

ASSESEMENT CRITERIA

- The documentation is created according to the design created in specific outcome 'Plan and design documentation for a computer program to agreed standards'.
- The documentation components created are created according to the plan specified in specific outcome 'Plan and design documentation for a computer program to agreed standards'.
- The documentation created is structured sensibly, defining how program specifications have been met.

Activity	Practical	SO	AC
2	**20. Describe how to create software documentation, (10)**	2	1-3

1. Define the Audience and ScopeIdentify readers: Tailor depth and technical jargon to developers, system administrators, or end-users.Select the type: Choose between internal docs (architecture, API references) or external docs (user manuals, installation guides).Determine boundaries: Map exactly which features, code modules, or workflows the document will cover.

2. Choose the Platform and ToolsVersion control repositories: Store documentation directly alongside the source code in systems like GitHub to track edits alongside code updates.Knowledge hubs: Use collaborative wikis like Atlassian Confluence or SharePoint for internal engineering and project guidelines.Static site generators: Deploy docs-as-code frameworks like Nextra or Docusaurus to convert Markdown files into public-facing websites.

3. Structure the ContentStandard overview: Introduce the system's core goals, architectural decisions, and the explicit business problems it solves.Setup instructions: Write clear, chronological prerequisites, installation steps, and system dependencies.Actionable examples: Embed minimal code snippets, visual flowcharts, and targeted screenshots.Troubleshooting matrix: Build an explicit table matching common error symptoms to their exact root causes and resolutions.

4. Review and MaintainApply style guides: Follow an established standard, like the Google Developer Documentation Style Guide, to keep formatting and tone consistent.Collaborative peer review: Pass drafts through technical experts for accuracy and non-technical testers for readability.Automate deployments: Connect documentation platforms to CI/CD pipelines to rebuild and republish guides automatically with every software release.Schedule audits: Regularly review the documentation database to catch outdated parameters, deprecated features, or broken external hyperlinks.

SPECIFIC OUTCOME 3:

Review documentation for a computer program for completeness.

ASSESEMENT CRITERIA

- The review identifies if documentation was created according to the design created in specific outcome 'Plan and design documentation for a computer program to agreed standards'.
- The review identifies if documentation was created consistent with the computer program being documented.

Activity	Question	SO	AC
3	**17. In pairs describe the importance of reviewing documentation for a computer program for completeness. (3)**	3	1-2

Reviewing software documentation for completeness ensures that no critical information, edge cases, or functional steps are missing, which helps prevent costly errors, reduce user confusion, and preserve accurate institutional knowledge for future maintenance.When working in pairs (such as in a driver/navigator setup similar to pair programming), partners can divide the review to maximize thoroughness:Partner A (The Technical Validator): Focuses on system and architectural documentation integrity. They verify that API endpoints, data flowcharts, and code logic align completely with the actual program.Partner B (The End-User/QA Advocate): Reviews user manuals, installation guides, and troubleshooting steps. They ensure no steps are skipped so that a beginner or third party can use or maintain the program without getting stuck.By cross-referencing their perspectives against established program documentation principles, the pair catches hidden gaps that a single reviewer might overlook.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
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4				
5				
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08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

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Assessor	**Date**	**Moderator**	**Date**

WEEK 11

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115374

Demonstrate an understanding of the use of web-sites in business

Unit Std #	115374
NQF Level	5
Notional hours	40
Credit(s)	4
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1 : Discuss the use of web sites in business.
• The discussion outlines the primary basis on which web sites are designed • The discussion identifies different uses of the Internet as business tool
Specific Outcome 2 : Demonstrate an understanding of the basic concepts of web-site design for business use.
• The demonstration identifies the basic concepts to apply to design and production of the web pages • The demonstration explains the use of techniques to make the web site accessible and popular • The demonstration recognises the need to protect a web site

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (4) x 10 = 40	12HRS	28HRS

1.	**Learning Programme Name:**			REFER TO COVER PAGE							
2.	**SAQA Qualification/ Unit Standard Title:**			REFER TO COVER PAGE							
3.	**Qualification/**		**4.**	**SAQA ID Number**		**5.**	**NQF Level**	5	**6.**	**Credits**	**4**
	Unit Standard										
7.	**PURPOSE for offering this programme to your learners:**			REFER TO NEXT PAGE							
8.	**TARGET AUDIENCE for this specific programme:**			REFER TO NEXT PAGE							
9.	**Entry/Admission Requirements:**			REFER TO NEXT PAGE							

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 12 hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =28 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
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Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

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- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

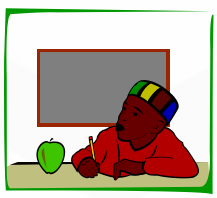
Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: Discuss the use of web sites in business.
ASSESEMENT CRITERIA
❖ The discussion outlines the primary basis on which web sites are designed ❖ The discussion identifies different uses of the Internet as business tool

Activity	Practical	SO	AC
1	**1.1 In small groups Discuss the use of web sites in business (10)**	1	1-2

Key Discussion Points for Small GroupsCredibility & First Impressions: Modern consumers expect legitimate businesses to have a professional website. A well-designed site establishes authenticity through customer testimonials, case studies, and team bios.24/7 Availability: Unlike brick-and-mortar stores with limited hours, a website operates as a constant salesperson, answering FAQs, processing transactions, and taking bookings around the clock.Marketing Hub & SEO: Search engine optimization (SEO) drives organic, long-term traffic. Every social media campaign or offline promotion should funnel back to a platform you fully own and control.

E-Commerce & Transactions: Websites enable digital storefronts (e-commerce) to sell products or services directly to a wider geographic market.Data & Analytics: Business websites allow you to track consumer behavior, optimize user journeys, and refine marketing strategies

SPECIFIC OUTCOME 2 : Demonstrate an understanding of the basic concepts of web-site design for business use.
ASSESEMENT CRITERIA
❖ The demonstration identifies the basic concepts to apply to design and production of the web pages ❖ The demonstration explains the use of techniques to make the web site accessible and popular ❖ The demonstration recognises the need to protect a web site

Activity	Question	SO	AC
2	**18. In pairs demonstrate an understanding of the basic concepts of web-site design for business use. (10)**	2	1-3

To demonstrate this understanding practically, business websites must incorporate several foundational elements:Clear Purpose & Call to Action (CTA): Define one primary goal per page (e.g., generating leads, making purchases) supported by clear, compelling CTAs.Mobile Responsiveness: Ensure sites function flawlessly across all devices using responsive design principles.Visual Consistency: Apply the company's distinct branding, typography, and color schemes.Security & Privacy: Protect user data to build vital trust with customers.Would you like to pick a specific business type (e.g., e-commerce, consulting, or a local restaurant) so we can map out a wireframe layout and user journey together

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
3				
4				
5				
6				

08 April 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR

I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115385

Demonstrate an understanding of the principles of implementing and managing an e-Commerce web site

Unit Std #	115385
NQF Level	5
Notional hours	120
Credit(s)	12
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1 : Identify various marketing options for use within e-Commerce web sites.
• The identification lists marketing options which are relevant to the web site • The identification describes how the marketing options could be used to enhance the web site.
Specific Outcome 2 : Explain the legal issues concerning the building of an e-Commerce web site.
• The explanation identifies legal issues concerning e-Commerce web sites • The explanation outlines how these legal issues should be incorporated when building the web site • The explanation ensures that all relevant legal issues have been addressed when implementing the web site
Specific Outcome 3 : Demonstrate an understanding of hosting arrangements for an e-Commerce web site
• Identify possible service providers for hosting the web site • Identify the costs involved in order to host the web site • Understand contractual agreements for hosting the web site
• **Specific Outcome 4 : Demonstrate an understanding of providing support for an e-Commerce web site.**
• The provision ensures that a "help facility" is available on the web site • The provision ensures that any technical difficulties experienced on the web site, can be resolved timeously and with ease • The provision ensures that telephonic or online (or similar) assistance is available in the event of difficulties being experienced.

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (12) x 10 = 120	36HRS	84HRS

1.	**Learning Programme Name:**			REFER TO COVER PAGE							
2.	**SAQA Qualification/ Unit Standard Title:**			REFER TO COVER PAGE							
3.	**Qualification/**		**4.**	**SAQA ID Number**		**5.**	**NQF Level**	5	**6.**	**Credits**	**12**
	Unit Standard										
7.	**<u>PURPOSE</u> for offering this programme to your learners:**			REFER TO NEXT PAGE							
8.	**<u>TARGET AUDIENCE</u> for this specific programme:**			REFER TO NEXT PAGE							
9.	**Entry/Admission Requirements:**			REFER TO NEXT PAGE							

10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 36 hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =84 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch			**Registration No:**	
Contact Details	**email:**			**@primeserv.co.za**
	Phone:		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

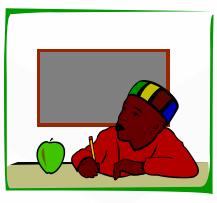
Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: **Identify various marketing options for use within e-Commerce web sites.**
ASSESEMENT CRITERIA
❖ The identification lists marketing options which are relevant to the web site ❖ The identification describes how the marketing options could be used to enhance the web site

Activity	Practical	SO	AC
1	**1.1 In pairs Identify various marketing options for use within e-Commerce web sites. (10)**	1	1-2

Core E-Commerce Marketing OptionsSearch Engine Optimization (SEO) & PPC: Experts on Quora emphasize combining organic rankings via SEO with paid search ads (such as Google Ads) to immediately target active shoppers.Email Marketing & Retargeting: Automated campaigns and retargeting ads across display and social networks bring back window shoppers and reduce cart abandonment, as outlined in the DHL E-Commerce Guide.Social Media & Influencer Marketing: Platforms like Facebook and Instagram excel at visual product promotion and direct social selling.Content Marketing & Affiliate Programs: Content marketing engages audiences with valuable information, while affiliate marketing partners with third parties for a commission on sales.Loyalty Programs: Rewarding repeat customers builds retention.

SPECIFIC OUTCOME 2 : Explain the legal issues concerning the building of an e-Commerce web site.
ASSESEMENT CRITERIA
❖ The explanation identifies legal issues concerning e-Commerce web sites ❖ The explanation outlines how these legal issues should be incorporated when building the web site ❖ The explanation ensures that all relevant legal issues have been addressed when implementing the web site

Activity	Practical	SO	AC
2	**21. In groups discuss the legal issues concerning the building of an e-Commerce web site. (10)**	2	1-3

Data Protection & Privacy: Group members should outline how user data (names, emails, payment info) is collected, stored, and processed. You will need a compliant Privacy Policy that respects regulations like the EU's GDPR or South Africa's POPIA.Consumer Protection & Contracts: E-commerce platforms must establish valid electronic transactions. Discuss how you will implement easily accessible Terms & Conditions that explicitly state pricing, delivery, warranties, and return policies. Under frameworks like the Electronic Communications and Transactions Act (ECTA), these digital contracts must be perfectly clear to be legally binding.Intellectual Property (IP): Address the risk of IP infringement. Emphasize that all graphics, product descriptions, logos, and software used on the site must be custom-made, properly licensed, or original to avoid copyright or trademark disputes.Payment & Transaction Security: The group needs to evaluate payment gateways. The site must meet Payment Card Industry Data Security Standard (PCI DSS) requirements to process financial details safely and avoid liability.Jurisdiction & Tax Compliance: Discuss which tax laws (such as VAT) apply based on where your buyers and sellers are located. You must also determine which regional or international legal jurisdiction governs your Terms of Service if disputes arise.

SPECIFIC OUTCOME 3:

Demonstrate an understanding of hosting arrangements for an e-Commerce web site.

ASSESEMENT CRITERIA

- Identify possible service providers for hosting the web site
- Identify the costs involved in order to host the web site
- Understand contractual agreements for hosting the web site

Activity	Question	SO	AC
3	**19. In pairs demonstrate an understanding of hosting arrangements for an e-Commerce web site. (5)**	3	1-3

1. Types of Hosting InfrastructureShared Hosting: Multiple websites share a single server 1.1.1. Cost-effective, best for startups or small shops with low traffic 1.1.1.VPS (Virtual Private Server): Server partitioned into dedicated virtual units 1.1.1. Provides more control and scalability for growing stores 1.1.1.Dedicated Hosting: An entire server leased solely to your store 1.1.1, 1.1.11. Maximum performance and security for high-volume operations 1.1.1.Cloud Hosting: Uses a network of virtual servers 1.1.1. Highly reliable, automatically adjusts to traffic spikes (e.g., Black Friday) 1.1.1.2. Service Providers & SupportLocal Providers (South Africa): Companies like xneelo and Domains.co.za offer localized infrastructure and support 1.1.17, 1.2.4.All-in-One Platforms: Services like Wix bundle the site builder, hosting, and security in one 1.1.6, 1.2.14.

Key Support Services: Ensure your partner provides CDN (Content Delivery Network) integration, 24/7 technical assistance, and automated backups.3. Costs & Contractual AgreementsMonthly Fees: Shared plans typically range from roughly $5–$20 or ~R189–R389 per month 1.1.6, 1.1.7. VPS or dedicated plans scale from $30 to over $250+ depending on traffic needs 1.1.6.Contractual Terms: Review uptime guarantees (usually 99.9%), renewal pricing, data usage limits, and Service Level Agreements (SLAs) before signing 1.1.2, 1.1.8, 1.2.15.If you and your partner are preparing a specific project, tell me:What type of products are you selling?What is your estimated monthly visitor traffic?

SPECIFIC OUTCOME 4:

Demonstrate an understanding of providing support for an e-Commerce web site.

ASSESEMENT CRITERIA

- The provision ensures that a "help facility" is available on the web site
- The provision ensures that any technical difficulties experienced on the web site, can be resolved timeously and with ease
- The provision ensures that telephonic or online (or similar) assistance is available in the event of difficulties being experienced.

Activity	Question	SO	AC
4	**8. In pairs Demonstrate an understanding of providing support for an e-Commerce web site (5)**	4	1-3

Providing robust e-Commerce support requires combining seamless front-end customer assistance with reliable back-end technical maintenance. Operating in pairs allows teams to divide these core responsibilities:Front-End / Customer-Facing Support: Partner A manages the human element by handling queries via Shopify live chat, emails, and phone assistance. They maintain a detailed Salesforce FAQ section, process returns, and track orders to ensure customer satisfaction.Back-End / Technical Support: Partner B manages the platform's stability by monitoring Pair Networks web hosting, securing TP payment gateways, fixing bugs, and updating product catalogs in real-time.If you'd like to prepare for an assessment or presentation, let me know:Would you like a script for a roleplay between the two partners?Do you need a checklist of technical issues to demonstrate troubleshooting?Should we focus on B2B or B2C e-commerce?

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
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4				
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08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I ……..... (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simnikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

WEEK 12

National Certificate: Information Technology (Systems Development):

SAQA 48872 -LEVEL 5- 131 CREDITS

LEARNER WORKBOOK

US ID: 115378

Demonstrate an understanding of advanced object-oriented programming

Unit Std #	115378
NQF Level	6
Notional hours	14
Credit(s)	140
Field	Field 03 - Physical, Mathematical, Computer and Life Sciences
Sub-Field	Information Technology and Computer Sciences
Qualification	National Certificate: Information Technology (Systems Development) LEVEL 5- SAQA- 48872- 131 CREDITS

Specific Outcome 1 : Demonstrate an understanding of inheritance in object oriented programming (OOP).
• The demonstration identifies situations from real-life problems where inheritance is applicable. The demonstration implements inheritance in OOP language.
Specific Outcome 2 : Demonstrate an understanding of polymorphism in object oriented programming (OOP).
• The demonstration identifies situations from real-life problems where polymorphism is applicable. • The demonstration implements polymorphism in OOP language.
Specific Outcome 3 : Analyse problem situations to plan an OOP implementation.
• The analysis describes basic UML standards to identify classes. • The analysis identifies base and sub-classes in a problem situation. • The analysis identifies polymorphism in a problem situation. • The analysis identifies inter-class communication in a problem situation
• **Specific Outcome 4 : Use abstract classes to optimise re-usability.**
• Describe the purpose of abstract classes in the optimisation of re-usability. • Describe wrapper classes and their need in OOP design. • Implement abstract and wrapper classes in OOP language.
• **Specific Outcome 5 : Use basic design patterns to optimise re-usability.**
• Describes different design patterns. • Explain the advantages of using the chosen design pattern. • Implement the chosen design pattern in an OOP language.
• **Specific Outcome 6 : Document the program according to industry standards**

• The documentation includes annotation of the program with a description of program purpose and design specifics. • The documentation includes the layout of the program code including indentation and other acceptable industry standards • The documentation includes full internal and external documentation, with a level of detail that enables other programmers to analyse the program • The documentation reflects the tested and implemented program, including changes made during testing of the program

NOTIONAL HOURS BREAKDOWN

The candidate undertaking this unit standard is best advised to at least spend one hundred hours of study time on this learning programme. Below is a table which demonstrates how these one hundred hours could be spread:

TIMEFRAME

Total Notional Hours	**Contact Time**	**Non contact-** **Self-Study/Assessment**
Credits (14) x 10 = 140	42HRS	98HRS

1.	**Learning Programme Name:**			REFER TO COVER PAGE							
2.	**SAQA Qualification/ Unit Standard Title:**			REFER TO COVER PAGE							
3.	**Qualification/**		**4.**	**SAQA ID Number**		**5.**	**NQF Level**	6	**6.**	**Credits**	**14**
	Unit Standard										

7.	**<u>PURPOSE</u> for offering this programme to your learners:**	REFER TO NEXT PAGE
8.	**<u>TARGET AUDIENCE</u> for this specific programme:**	REFER TO NEXT PAGE
9.	**Entry/Admission Requirements:**	REFER TO NEXT PAGE
10.	**Timeframe for Training:** **(Total Hours/Days/ Weeks)**	**Theory content** –Role play, Simulation, Group work, Pair work = 42 hrs. **Non contact session**- self-study, assignment, practise guided by coach or mentor, formative assessment and summative assessment =98 hrs.

Contact Details

Standard Title:	

Course:	

Assessor Details				
Name				
Branch		**Registration No:**		
Contact	**email:**		**@primeserv.co.za**	

Details	**Phone:**		**Fax:**	
Moderator Details				
Name				
Branch				
Contact Details	**email:**		**Registration No:**	
	Phone:		**Fax:**	
Candidate Details				
Name		**ID No**		
Job		**Company**		
Department				
Contact Details	**Email:**			
	Phone:		**Fax:**	

I confirm that:

- the purpose of the assessment has been clearly explained to me
- the criteria have been discussed with me, and I know I will be assessed against these criteria
- I know when and where I will be assessed, and I was given fair notice
- I know how the assessment will be done, and any other requirements related to the assessment

Signed: _____ **Date:** _____

Overall Assessment Decision	Competent	Not yet competent	
Candidate's Signature		**Date**	
Assessor's Signature		**Date**	
Moderator's Signature		**Date**	

Control sheet

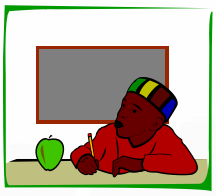
Exercise No	Date completed	Learner's signature	Mentor's signature	Moderator's signature

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Self-check No	Date completed	Learner's signature	Mentor's signature	Moderator's signature
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4				
5				

Reflection	Date completed	Learner's signature	Mentor's signature	Moderator's signature
1				

Answer all questions in this guide



SPECIFIC OUTCOME 1: Demonstrate an understanding of inheritance in object oriented programming (OOP).
ASSESEMENT CRITERIA
❖ The demonstration identifies situations from real-life problems where inheritance is applicable. ❖ The demonstration implements inheritance in OOP language.

Activity	Practical	SO	AC
1	**1.1 In pairs describe the Fundamental Principles of Object-Oriented Programming (OOP)(5)** **1.2 Outline different Types of inheritance and their application (8)**	1	1-2

Object-Oriented Programming (OOP) is built on four core pillars (as there is a clear consensus) that organize code into modular, reusable, and secure units called objects:1. EncapsulationWhat it is: Bundling data (variables/attributes) and the methods (functions/behaviors) that operate on that data into a single unit (a class).How it works: It hides the internal state of an object from the outside world and restricts direct access. Interactions happen purely through controlled public interfaces (like methods or getters/setters).Real-World Pair Analogy: A capsule medicine or a locked safe. You don't need to know how the chemicals work internally; you just swallow the capsule, or you can only change the contents if you have the key (methods).

2. AbstractionWhat it is: Hiding complex implementation details and only showing the essential, relevant features of an object to the user.How it works: It allows developers to focus on what an object does rather than how it does it, reducing programming complexity.Real-World Pair Analogy: Driving a car. You press the accelerator or turn the steering wheel to drive, but you do not need to know the intricate internal mechanics of the engine or fuel injection to make it move.3. InheritanceWhat it is: A mechanism that allows a new class (child/subclass) to adopt the properties and methods of an existing class (parent/superclass).How it works: It promotes code reusability and creates a clear, hierarchical "is-a" relationship. The child class can reuse the parent's logic while adding or overriding its own unique behaviors.

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SPECIFIC OUTCOME 2 :

Demonstrate an understanding of polymorphism in object oriented programming (OOP).

ASSESEMENT CRITERIA

- The demonstration identifies situations from real-life problems where polymorphism is applicable.
- The demonstration implements polymorphism in OOP language.

Real-World Pair Analogy: A parent and a child. A child inherits physical traits and genetic features from the parent, but may also develop their own distinct skills and traits.

Activity	Practical	SO	AC
2	**22. Demonstrate an understanding of polymorphism in object oriented programming (OOP). (5)**	2	1-2

Core Types of PolymorphismOOP languages generally implement polymorphism in two distinct ways:1. Runtime Polymorphism (Dynamic Binding)This occurs when the specific method implementation to execute is determined at program execution (runtime) rather than compilation. It is achieved via Method Overriding, where a subclass provides its own custom logic for a method already defined in its parent class.

Mechanism: A collection or variable is typed to a broad superclass, but the program dynamically triggers the subclass implementation.Real-world Example: A Payment class has a process() method. The derived subclasses CreditCardPayment and CryptoPayment both implement process(), but one charges a merchant bank while the other broadcasts a blockchain transaction.

2. Compile-time Polymorphism (Static Binding)This occurs when the compiler determines exactly which method to execute before the program runs. It is achieved via Method Overloading, where multiple methods inside the same class share an identical name but maintain different parameter structures (signatures).Mechanism: The compiler differentiates the methods by analyzing the number, order, or data types of the arguments provided.Real-world Example: A Notification class might have a send(String message) method and a overloaded send(String message, String attachmentPath) method.

SPECIFIC OUTCOME 3:

Analyse problem situations to plan an OOP implementation.

ASSESEMENT CRITERIA

- The analysis describes basic UML standards to identify classes.
- The analysis identifies base and sub-classes in a problem situation.
- The analysis identifies polymorphism in a problem situation.
- The analysis identifies inter-class communication in a problem situation.

Activity	Question	SO	AC
3	**20. Define the Role of UML in OO design: (5)** **21. Define System Classes (2)**	3	1-4

The Unified Modeling Language (UML) serves as the standard visual blueprint used to specify, visualize, construct, and document the architecture of object-oriented (OO) software systems. It bridges the gap between conceptual requirements and concrete programming code by converting abstract OO concepts into standard graphical models.

Key Roles in OO DesignStructural Visualization: UML provides structural diagrams—most notably Class Diagrams—to define the static skeleton of an application, capturing classes, properties, behaviors, and inheritance.Behavioral Modeling: It uses dynamic diagrams like Sequence and State Machine Diagrams to map out object interactions, message ordering, and runtime workflows.Encapsulation & Scope Control: UML visually enforces data hiding by using visibility notations such as private (-), public (+), and protected (#) directly inside model partitions.

Standardized Collaboration: It provides a universal syntax that acts as a "Rosetta stone," allowing developers, architects, and non-technical stakeholders to communicate cleanly without looking at code.

Definition of System ClassesIn object-oriented software design, System Classes are specialized, low-level classes responsible for software management, control, and core computing infrastructure. Unlike business domain classes (which model real-world concepts like Customer or Order), system classes provide the underlying mechanisms that allow the software application to execute, communicate, and interact with its computing environment safely.

SPECIFIC OUTCOME 4:

Use abstract classes to optimise re-usability.

ASSESEMENT CRITERIA

- ❖ Describe the purpose of abstract classes in the optimisation of re-usability.
- ❖ Describe wrapper classes and their need in OOP design.
- ❖ Implement abstract and wrapper classes in OOP language.

Activity	Question	SO	AC
4	**9. What are Wrapper classes? (2)** **10. Outline the Purpose of abstract classes in the optimisation of re-usability (5)**	4	1-2

Wrapper classes in programming (predominantly discussed in Java) provide a mechanism to encapsulate primitive data types—such as int, char, boolean, and double—into objects. This transformation enables primitives to participate in object-oriented operations where standard data values are not accepted.Why Wrapper Classes ExistObject-Oriented Frameworks: Data structures like ArrayList or HashMap exclusively store objects and reject primitive types. Wrappers allow numerical and true/false values to populate these collections.Utility Methods: They provide static helper functions—such as Integer.parseInt() for converting text to numbers, or maximum and minimum constants.Null Support: Unlike primitives (which must always hold a value like 0 or false), wrapper objects can represent a state of null

Primitive vs. Wrapper TypesEvery primitive data type has a corresponding, capitalized equivalent defined in the java.lang package:Primitive TypeWrapper ClassintIntegercharCharacterdoubleDoublebooleanBoolean

Boxing and UnboxingTo transition between the two representations, the language relies on two fundamental operations:Boxing: Converting a primitive into its corresponding wrapper object (e.g., changing int to Integer).Unboxing: Extracting the raw primitive value back out of the wrapper object.

SPECIFIC OUTCOME 5:

Use basic design patterns to optimise re-usability.

ASSESEMENT CRITERIA

- Describes different design patterns.
- Explain the advantages of using the chosen design pattern.
- Implement the chosen design pattern in an OOP language.

Activity	Question	SO	AC
5	**11. Identify different types of design patterns and list the advantages of and implement the chosen design pattern. (10)**	4	1-3

Categories of Design PatternsCreational Design Patterns: Deal with object creation mechanisms, optimizing how and when objects are instantiated (e.g., Singleton, Factory Method, Builder).Structural Design Patterns: Explain how to assemble classes and objects into larger, flexible structures (e.g., Adapter, Decorator, Facade).Behavioral Design Patterns: Focus on communication and the assignment of responsibilities between objects (e.g., Observer, Strategy, State).

Chosen Pattern: The Factory Method PatternThe Factory Method pattern belongs to the creational category. It provides an interface for creating objects in a superclass, but allows subclasses to alter the type of objects that will be created.AdvantagesLoose Coupling: Code interacts only with the abstract interface or base class rather than concrete implementations.Extensibility: New classes and subclasses can be introduced easily without breaking existing core code, adhering to the SOLID principle of being open for extension but closed for modification.Single Responsibility: Object creation logic is centralized, preventing code duplication across the system.

SPECIFIC OUTCOME 6:

Document the program according to industry standards.

ASSESEMENT CRITERIA

- ❖ The documentation includes annotation of the program with a description of program purpose and design specifics.
- ❖ The documentation includes the layout of the program code including indentation and other acceptable industry standards
- ❖ The documentation includes full internal and external documentation, with a level of detail that enables other programmers to analyse the program
- ❖ The documentation reflects the tested and implemented program, including changes made during testing of the program

Activity	Question	SO	AC
6	**4. In pairs describe how to Document the program according to industry standards (10)**	6	1-4

1. Source Code & Low-Level DocumentationInline Comments: Use comments sparingly to explain the "why" behind complex algorithms, rather than the "what".Docstrings: Write structured explanations at the top of classes and methods defining parameters, return types, and exceptions. Industry tools like Javadoc or Sphinx automate this.Conventions: Follow a strict, team-wide naming convention for variables and functions to make code self-documenting.2. High-Level & Project DocumentationREADME Files: Create a root-level markdown file containing project descriptions, installation prerequisites, and a quick-start example.Architecture Docs: Maintain diagrams and technical design overviews detailing data flow and system structure.API Documentation: If applicable, publish endpoint specifications using open standards like Swagger/OpenAPI.

3. User & Operational DocumentationEnd-User Manuals: Write step-by-step guides and tutorials catering to non-technical end-users or system administrators.Version Control & Maintenance: Host documentation alongside code in repositories like GitHub to enforce version control. Establish a schedule using IT Documentation Best Practices to review and update outdated documents.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply

2. Reasonable compliance (Not acceptable for final evaluation)

3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1				
2				
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4				
5				
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08 July 2026

S.Picane

Learner Signature **Date**

Facilitators' Signature **Date**

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met**	**Requirements not met**
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I …….. (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, Simmikiwe Picane .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

